

Hail Mary Pass: Contests with Stochastic Progress

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Abstract

This paper studies equilibrium behavior in contests with stochastic progress. Participants have access to a safe action that yields deterministic progress, but they can also take a risky move that stochastically affects their progress toward the goal, and hence their relative position. In the unique *well-behaved* Markov perfect equilibrium of this dynamic contest, the follower drops out if the leader establishes a substantial lead, but resorts to “Hail Marys” beforehand: no matter how low the expected return of the risky move, the follower takes it in an attempt to catch up. Moreover, if the risky move has a medium return (between high and low), the leader also adopts it when the follower is close to dropping out, reflecting an unexpected preemptive motive. We also analyze how this equilibrium behavior shapes the optimal prize allocation.

Keywords: Dynamic contests, Markov perfect equilibrium, contest design

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1 Introduction

This paper studies contests with stochastic progress induced by risky moves. Agents exert effort and compete to get ahead. They can take a safe action that delivers deterministic progress, or a risky move that stochastically affects their relative position. The central question is how such risky moves shape equilibrium behavior. Specifically, under what circumstances do agents take them? Will they adopt a risky move even when its expected return is arbitrarily low?

In the baseline model (Section 2), two agents compete for a prize. Progress is publicly observed, and each agent decides *whether and how* to exert effort. The agents are identical except for their initial positions. If an agent chooses not to exert effort, he irreversibly drops out of the contest. Otherwise, he incurs a flow cost and selects either the *safe action*, which makes progress deterministically at a constant rate, or the *risky move*, which stochastically affects his progress. The current leader wins at a Poisson rate, ending the contest. The contest is “winner-takes-all”: the entire prize goes to the winner at the end of the contest.¹ Agents play a pure-strategy *Markov perfect equilibrium (MPE)*. In our model, a Markov strategy reduces to a *stopping rule*, which specifies *whether* to exert effort, and an *action rule*, which specifies *how* to exert effort.

Our main result is that the follower drops out once the leader establishes a substantial lead, but resorts to “Hail Marys” beforehand: no matter how low the expected return of the risky move, the follower takes it in an attempt to catch up. Moreover, if the risky move has a *medium* return (between *high* and *low*), the leader also adopts it when the follower is close to dropping out, reflecting an unexpected preemptive motive.

Toward this result, we first establish three types of MPEs of the dynamic contest and characterize the structure of equilibrium strategies (Section 3). In each type of MPE, the follower takes the risky move regardless of its expected return; the leader’s behavior, by contrast, depends on how the risky move compares to the safe action. The analysis splits into three cases, according to whether the risky move has a *low*, *medium*, or *high* return. When the return is low, the leader never finds it optimal to take the risky move, and the MPE features each agent taking the risky move if and only if he is the follower (Proposition 1). In the opposite case where the return is high, both agents optimally take the risky move, since it is a worthwhile investment (Proposition 2). Surprisingly, when the return is medium, i.e., between the previous two cases, the leader also takes the risky move, but only when the follower is close to dropping out (Proposition 3). We further show that these MPEs are unique within a broad and natural family of MPEs, which we call *well-behaved* MPEs (Propositions 1’, 2’, and 3’).

A natural next question is how such equilibrium behavior affects the optimal prize allocation. We

¹In Section 4, we extend the baseline model to allow the follower to win at a positive Poisson rate and the loser to receive a positive prize.

address this question by extending the baseline model to allow the designer to allocate prizes subject to a fixed budget (Section 4), while preserving the derivations and intuitions developed in the baseline model. We show that three natural objectives for the designer are equivalent in our contest (Proposition 4). Furthermore, the solution to the designer’s problem shows that “winner-takes-all” is optimal only when the designer’s budget is small, and otherwise the optimal allocation rewards the loser as well (Proposition 5).

Related Literature This paper is related to the large literature on innovation and patent races, pioneered by Loury (1979) and Dasgupta and Stiglitz (1980). Within this literature, two main outcomes are emphasized: (i) ε -preemption (Fudenberg et al., 1983; Harris and Vickers, 1985a,b; Lippman and McCardle, 1988), where a slight advantage causes the opponent to drop out immediately, and (ii) *increasing dominance* (Grossman and Shapiro, 1987; Harris and Vickers, 1987; Lippman and McCardle, 1987), where the follower falls further behind and the leader builds up its advantage. We introduce risky moves into a contest of this kind and examine their impact on equilibrium behavior. The follower stays in the contest when not far behind, hoping to use a “Hail Mary” to catch up. Moreover, the equilibrium degenerates to ε -preemption as the variance of the risky move vanishes, showing that risk-taking is essential to the result.

Our model is closest to Anderson and Cabral (2007), who also study dynamic competition in which the agents’ choices affect the variance of a state variable. In their model, players influence the variance but not the mean, and there is no endogenous exit from the contest. In our model, the agents’ actions can affect both the mean and the variance, and agents can drop out at any time. As a result, the two papers obtain different forms of MPEs. In particular, the mean return of the risky move, relative to the safe action, determines whether the leader engages in risk-taking. This channel does not arise in Anderson and Cabral (2007)’s setup, where only the variance is affected.

Technically, our model is a one-dimensional stochastic differential game, so it is naturally connected to the theoretical literature on such games (Girsanov, 1961; Gusein-Zade, 1969; Friedman, 1972; Pliska, 1973; Harris, 1993). Equilibrium existence is known to be delicate when the players’ actions can affect the variance of the state. Fortunately, in our model we are able to construct closed-form MPEs directly, thereby sidestepping the existence problem. A similar difficulty arises for uniqueness, and we have so far only shown that the constructed MPEs are unique within a broad and natural family of MPEs, which we call *well-behaved* MPEs. These are MPEs whose value functions satisfy the Hamilton-Jacobi-HJB equations of the contest game.

This paper is also related to a recent wave of research on contest design that focuses on feedback in contests (Yildirim, 2005; Ederer, 2010; Halac, Kartik, and Liu, 2017; Bimpikis, Ehsani, and Mostagir, 2019;

Benkert and Letina, 2020; Ely et al., 2022). Our model abstracts away from information disclosure to focus on the role of risky moves. Combining the two is a natural direction for future work.

The rest of the paper is organized as follows. Section 2 lays out the baseline model. The first main part, Section 3, establishes three types of MPEs and characterizes the structure of equilibrium strategies. The second main part, Section 4, then analyzes the impact of equilibrium behavior on the optimal prize allocation. Section 5 concludes. Appendix A contains the proofs of all results in the main text.

2 Model

The baseline model is a continuous-time contest with stochastic progress. Two agents (he) $i \in \{1, 2\}$ participate in a contest to compete for a prize of value $P > 0$, and discount at a common rate r .

At each instant $t \in [0, \infty)$, each agent observes their *progress* $(k_1(t), k_2(t))$ and decides *whether and how* to exert effort. The agents are identical except for their initial position $k_i(0)$. If an agent chooses not to exert effort, he irreversibly drops out of the contest. Otherwise, if agent i exerts effort, he incurs flow cost at rate c , and selects an action $a_i(t) \in \{0, 1\}$. The *safe action* $a_i = 0$ makes progress deterministically at a constant rate normalized to 1. The *risky move* $a_i = 1$ stochastically affects agent i 's progress, with mean return $1 + \pi$ and variance σ^2 . If both agents take risks simultaneously, we assume the risks are independent. That is, agent i 's progress up to time t evolves according to

$$dk_i = (1 + a_i\pi)dt + a_i\sigma dw_i,$$

where w_1, w_2 are independent Wiener processes.²

Assume that, while both agents exert effort, the current *leader* succeeds with Poisson rate λ , at which point the contest ends. That is, conditional on not having succeeded by time t , the leader's effort for an additional duration dt yields success during the interval $(t, t + dt)$ with probability λdt . Away from ties, the leader is the agent with greater progress: agent i succeeds at rate λ when $k_i > k_j$ and at rate zero when $k_i < k_j$. At a tie, we adopt a fair one-sided continuation convention: before any payoff-relevant event occurs, the game proceeds as if k_i were infinitesimally above k_j or infinitesimally below it, each with probability one half.³

²If one wants progress to be nonnegative, it is equivalent to view $k_i = \log K_i$ as the logarithm of progress, and change the law of motion to $dK_i/K_i = (1 + a_i(\pi + \frac{1}{2}\sigma^2))dt + a_i\sigma dw_i$.

³This convention serves only to define continuation at the knife-edge state $k_i = k_j$; the action and success rate assigned exactly at a tie are payoff-irrelevant, and the convention affects neither equilibrium behavior nor payoffs in the equilibria analyzed below.

In the baseline model, the contest is *winner-takes-all*: the entire prize P is awarded to the winner at the end of the contest.⁴ Agent i 's ex post payoff is therefore

$$e^{-r\tau}P \cdot \mathbf{1}(i \text{ succeeds at } \tau) - \int_0^{\tau_i} e^{-rt}c \, dt = e^{-r\tau}P \cdot \mathbf{1}(i \text{ succeeds at } \tau) - \frac{c}{r}(1 - e^{-r\tau_i}), \quad (1)$$

where τ is the realized contest duration and $\tau_i \leq \tau$ is the time at which agent i ceases to exert effort, either because he drops out or because the contest ends. If one agent drops out while the other is still exerting effort, the contest continues with the latter as the sole contestant: he succeeds with Poisson rate λ and incurs the flow cost c while he exerts effort.⁵ Since his progress no longer affects his payoff, his choice between the safe and risky action is payoff-irrelevant and is suppressed. Should he drop out before succeeding, the contest ends with no prize awarded.

We focus on (pure strategy) *Markov perfect equilibria* (MPE). Following Maskin and Tirole (2001), we define *Markov strategies* as those that depend only on the *payoff-relevant state*; that is, strategies that are measurable with respect to the *maximally coarse consistent partition* (the *Markov partition*) of histories. Let $\Delta k \equiv k_i - k_j$ denote the difference in the agents' progress. Applying the *affine invariance criterion*⁶ for the Markov partition (Maskin and Tirole, 2001, Theorem 3.3), one can show that Δk is indeed the payoff-relevant state of the contest game. A Markov strategy $(\hat{k}_i, s_i)$ for agent i therefore depends only on Δk , and consists of two parts: a *stopping rule*, which specifies *whether* to exert effort through the cutoff $\Delta k \leq -\hat{k}_i$,⁷ and an *action rule* $s_i(\cdot) : (-\hat{k}_i, \infty) \rightarrow \{0, 1\}$, which specifies *how* to exert effort. An MPE $(k^*, s^*) = (k_i^*, s_i^*, k_j^*, s_j^*)$ is a subgame-perfect equilibrium in which both players use Markov strategies.

In the next section, we will establish three types of MPEs of the dynamic contest in closed form, and show that the structure of the equilibrium strategies depends crucially on the return of the risky move relative to the safe action. We will also establish uniqueness within a reasonably large family of MPEs, which we call *well-behaved* MPEs.

We accordingly impose the HJB equations on the two open regions $k_i > k_j$ and $k_i < k_j$, joining the value functions at $k_i = k_j$ by the appropriate matching conditions.

⁴Later in Section 4, we extend the model in two ways. First, we allow the loser to also have a positive prize: the winner gets $\bar{P}$ and the loser gets $\underline{P}$, with $\bar{P} \geq \underline{P} \geq 0$. Second, we allow the follower to also have a positive hazard rate of success, $\bar{\lambda}$ when leading and $\underline{\lambda}$ when following, with $\bar{\lambda} > \underline{\lambda} \geq 0$ and each agent equally likely to lead at a tie.

⁵In the extended model, the sole remaining agent is treated as the leader: he succeeds at rate $\bar{\lambda}$, incurs the flow cost c while exerting effort, and receives $\bar{P}$ upon success.

⁶This requires verifying that the continuation utility functions are appropriate affine transformations of one another, together with some non-degeneracy conditions (e.g., in any period, each player can choose different actions so as to affect the opponent's decision problem thereafter).

⁷The cutoff representation is without loss. Lemma A.1 shows that in any MPE the set of states at which dropout is optimal is downward closed: if dropout is optimal at some Δk , it is optimal at every lower state. Under the convention that the agent drops out whenever indifferent, this stopping set is represented by a cutoff $-\hat{k}_i$, so agent i stays in the contest iff $\Delta k > -\hat{k}_i$. Assigning the boundary to dropout is only a tie-breaking convention and does not affect payoffs.

3 Three Types of Equilibria

In this section, we demonstrate three types of MPEs of the dynamic contest and characterize the structure of the equilibrium strategies. In each type of MPE, the follower drops out if the leader takes a substantial lead, but engages in “Hail Marys” before then: no matter how low the return of the risky move is, the follower undertakes it to try to catch up.

Moreover, the return of the risky move compared to the safe action is critical in determining the leader’s behavior. The analysis boils down to three cases, depending on whether the risky move has a *low*, *medium*, or *high return*. If the risky move has a low return, then the leader never finds it optimal to take it, and the MPE features that each agent takes the risky move if and only if he is the follower. In the opposite case where the risky move has a high return, both agents optimally bear risk as it is a worthwhile investment. Surprisingly, if the risky move has a medium return, i.e., in between the previous two cases, the leader also adopts it when the follower is close to dropping out, reflecting a preemptive motive.

We denote by $\phi \equiv \lambda P/c$ the *profitability* of the contest, which is the ratio between effective prize and flow cost. It crucially affects equilibrium behavior. Note that ϕ needs to be larger than 1 for there to be a meaningful equilibrium, because if $\lambda P < c$, even the leader cannot recover the cost and drops out immediately. We are now ready to formally define the three cases, i.e., *low/medium/high return*.

Definition 1 (Low/medium/high return).

1. *The risky move has a low return if*

$$\pi \leq 0.$$

2. *The risky move has a medium return if*

$$0 < \frac{\pi}{\sigma} < \frac{\sqrt{r+\lambda}}{2} \cdot f(\phi).$$

3. *The risky move has a high return if*

$$\frac{\pi}{\sigma} \geq \frac{\sqrt{r+\lambda}}{2} \cdot f(\phi).$$

Here $f(\cdot)$ is a strictly increasing function, ranging from 0 to 1 as ϕ rises from 1 to infinity.⁸

⁸The closed-form expression of $f(\cdot)$ is as follows:

$$f(\phi) = \frac{\sqrt{2\phi(\sqrt{\phi^2+8}+\phi)}-8}{\sqrt{\phi^2+8}+\phi}.$$

In other words, low return means that the risky move has a lower mean return compared with the safe action; medium return means that the risky move has a strictly higher mean return, but the difference is below a certain upper bound; high return means that the difference is above that upper bound.

In the remainder of this section, we show that there exists a different form of MPE in each of the three cases (Propositions 1, 2 and 3). Furthermore, these MPEs are unique within a reasonably large family of MPEs, which we refer to as *well-behaved* MPEs (Propositions 1', 2' and 3').

3.1 Equilibrium with Low Return: “Hail Mary Pass”

We begin our analysis with the low return case, where the risky move has a lower mean return compared with the safe action ($\pi \leq 0$). In the candidate MPE, the follower drops out if the leader establishes a substantial lead, but engages in “Hail Marys” before then. That is, no matter how bad the risky move is, the follower undertakes it in an attempt to catch up. In contrast, the leader always selects the safe action, but is prepared to switch to the risky move once caught up by the follower.

Proposition 1 formally states this MPE.

Proposition 1 (Low return case). *If the risky move has a low return, then there exists a unique $k^* > 0$ such that the following strategy profile constitutes an MPE:*

$$\text{Drop out iff } \Delta k \leq -k^*, \text{ and otherwise } s_i^*(\Delta k) = 1 \text{ iff } \Delta k < 0.$$

Proof. All proofs of the results in the main text are in Appendix A. □

The proof of Proposition 1 takes two steps. The first step is to convert the Hamilton-Jacobi-Bellman (HJB) equations into second-order ordinary differential equations using Itô’s lemma. The second step imposes the boundary conditions of value matching and smooth pasting, and verifies that the agents have no profitable deviations.

Figure 1 visualizes this MPE from the perspective of agent i . There are four regions: in the leftmost region, agent i drops out because he falls too far behind; conversely, the opponent agent j drops out in the rightmost region. In the two middle regions, blue indicates that agent i chooses to take risk, while red indicates that the opponent chooses to take risk. As shown, agent i takes the risky move if and only if he falls behind, which mirrors the “Hail Mary pass” situation.

Note that the MPE in Proposition 1 is completely specified by the stopping boundary k^* , which enables the comparative statics with respect to the modeling parameters. We summarize the results as follows:

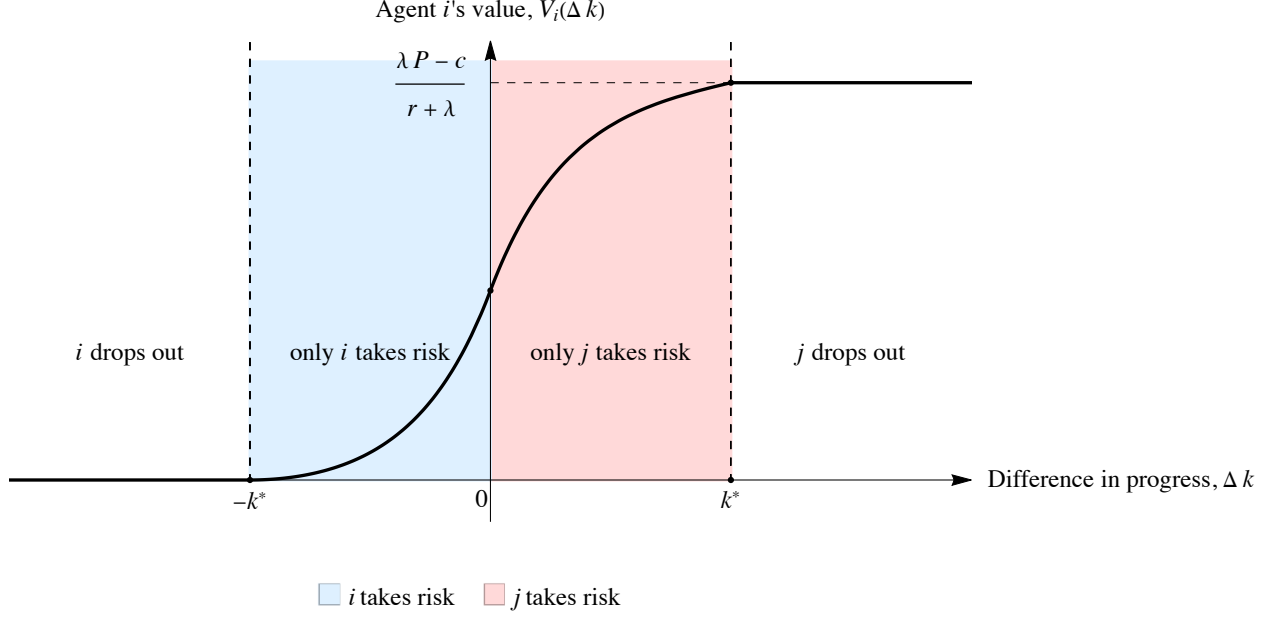


Figure 1: Equilibrium with low return: “Hail Mary pass”

1. The continuation region becomes larger as the contest becomes more profitable.

Formally, the stopping boundary k^* is strictly increasing in the ratio P/c . Moreover, in the limit case, k^* tends to infinity as either the prize P tends to infinity, or the flow cost c vanishes.

2. The continuation region becomes smaller as the risky move becomes less desirable.

The stopping boundary k^* is strictly increasing in the ratio π/σ (which by assumption takes a negative value in the low return case). Moreover, in the limit case, k^* tends to zero as either π tends to negative infinity, or σ vanishes. The only reason the follower chooses to stay is to rely on the “Hail Mary pass” to catch up, and the MPE degenerates to ε -preemption⁹ if that move is effectively removed.

3. The result is ambiguous if success occurs at a higher rate.

The stopping boundary k^* changes non-monotonically in the hazard rate of success λ , because two competing forces coexist. On the one hand, the contest becomes more profitable (ϕ increases) as λ increases, pushing k^* up. On the other hand, the follower becomes less likely to catch up in time, pushing k^* down. The first force dominates when λ is small, and the second force dominates when λ is large, demonstrating a single-peaked relationship overall.

⁹ ε -preemption is the equilibrium outcome in many standard models of innovation or patent races (Fudenberg et al., 1983; Harris and Vickers, 1985a,b; Lippman and McCardle, 1988), where a slight advantage causes the opponent to dropout immediately.

3.2 Equilibrium with High Return

The opposite case is easy to characterize, but perhaps not as interesting. When the risky move has a high return, both agents optimally bear the risk, because it is a worthwhile investment. The formal statement is Proposition 2.

Proposition 2 (High return case). *If the risky move has a high return, then there exists a unique $k^* > 0$ such that the following strategy profile constitutes an MPE:*

$$\text{Drop out iff } \Delta k \leq -k^*, \text{ and otherwise } s_i^*(\Delta k) = 1.$$

Figure 2 visualizes this MPE from the perspective of agent i . Consistent with Figure 1, blue indicates that agent i chooses to take risk, while red indicates that the opponent chooses to take risk. Both agents choose to take risk in this MPE, and therefore the entire middle region is purple.

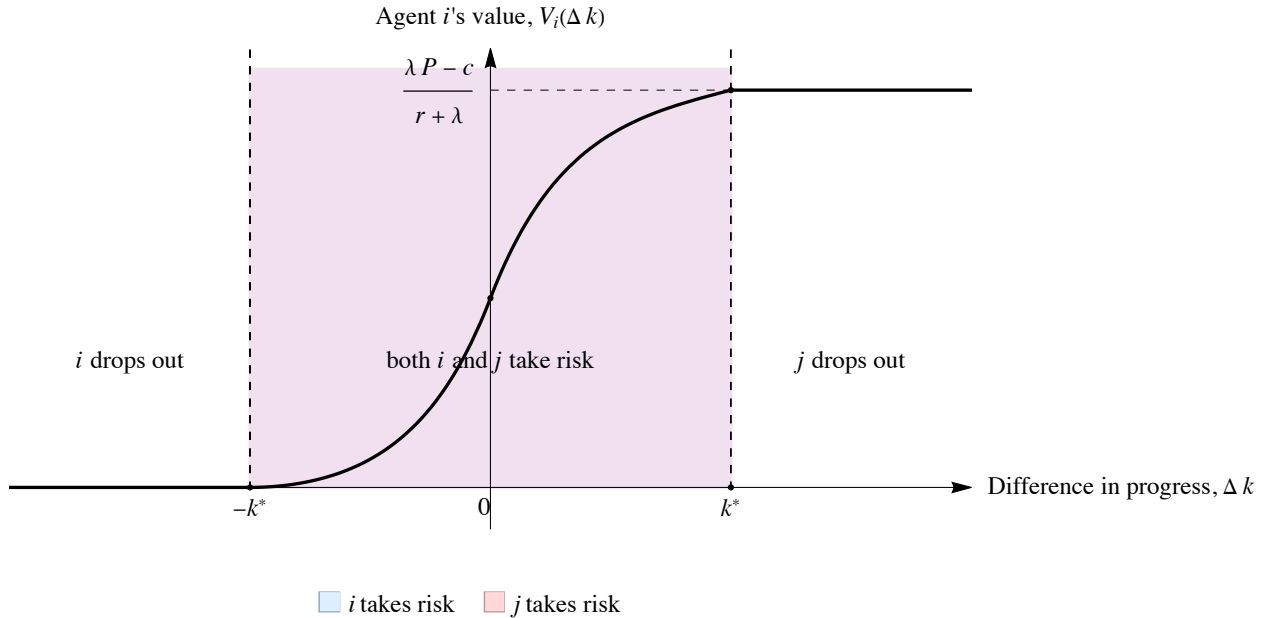


Figure 2: Equilibrium with high return

3.3 Equilibrium with Medium Return: “Hail Mary Pass” + Preemption

Finally, we analyze the remaining medium return case, where the risky move has a higher mean return than the safe action, yet not high enough to be a worthwhile investment in the absence of the opponent.

In the candidate MPE, not only does the follower take the risky move to try to catch up, but the leader also adopts it when the follower is close to dropping out. The formal statement is Proposition 3.

Proposition 3 (Medium return case). *If the risky move has a medium return, then there exists a unique pair of $k^* > k^{**} > 0$ such that the following strategy profile constitutes an MPE:*

$$\text{Drop out iff } \Delta k \leq -k^*, \text{ and otherwise } s_i^*(\Delta k) = 1 \text{ iff } (\Delta k < 0 \text{ or } \Delta k \geq k^{**}).$$

The MPE strategies outlined in Proposition 3 are combinations of the low and high return cases, and converge appropriately in the limit.¹⁰ The analysis reveals two motives for taking the risky move: one is the previous *Hail Mary motive*, which the follower use to try to catch up; the other is a new *preemptive motive*, where the leader seeks to drive the follower out. The preemptive motive is strongest at the follower's stopping boundary, because the temptation to win the contest for sure is especially strong for the leader, and the large lead makes it very unlikely that the follower will catch up even if the risky move does not work well.

Figure 3 visualizes this MPE from the perspective of agent i . Consistent with Figures 1 and 2, blue indicates that agent i chooses to take risk, while red indicates that the opponent chooses to take risk. The purple regions indicate that both agents take risk simultaneously, but for different reasons.

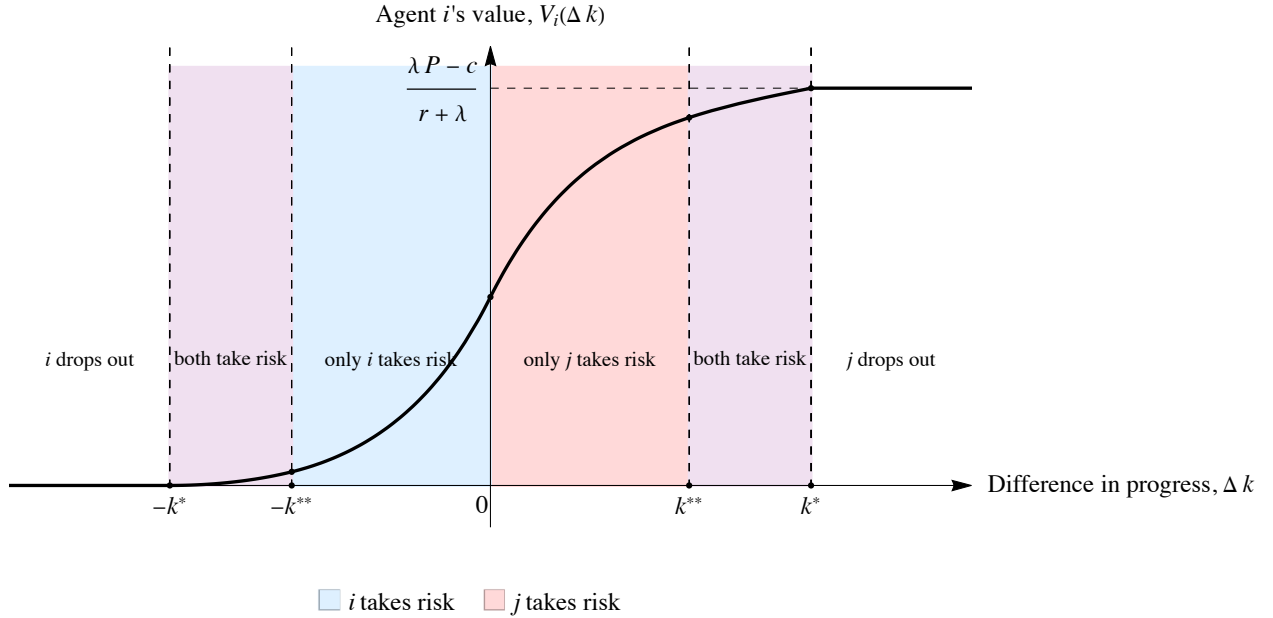


Figure 3: Equilibrium with medium return: “Hail Mary pass” + Preemption

¹⁰Formally, $k^* - k^{**} \rightarrow 0$ as $\frac{\pi}{\sigma} \downarrow 0$, and the MPE converges to the case of low return. Moreover, $k^{**} \rightarrow 0$ as $\frac{\pi}{\sigma} \uparrow \frac{\sqrt{r+1}}{2} \cdot f(\phi)$, and the MPE converges to the case of high return.

3.4 Uniqueness of the Three Types of Equilibria

So far, we have identified three types of equilibria, with two motives for taking the risky move. A natural question is: Are these equilibria (essentially) unique within their respective case (low/medium/high)?

In this subsection, we show that the answer is largely yes: the previously constructed equilibria are unique within a reasonably large family of MPEs. Formally, call an MPE *well-behaved* if it is a solution to the HJB equations of the contest game. We begin by outlining two key observations that help characterize any well-behaved MPE.

Suppose that the agents play according to some MPE $(k^*, s^*) = (k_i^*, s_i^*, k_j^*, s_j^*)$. Let $V_i(\Delta k)$ denote agent i 's value function in the continuation region $-k_i^* < \Delta k < k_j^*$, where both agents choose to exert effort and stay in the contest.

The first observation pins down the follower's behavior: he always takes the risky move as long as he stays in the contest.

Lemma 1. *In any MPE, the follower always takes the risky move if he chooses to stay in the contest. Formally, $s_i^*(\Delta k) = 1$ for all $(-k_i^*, 0)$.*

To understand the intuition behind Lemma 1, suppose that the follower instead chooses to stay in the contest and play it safe. One response available to the leader is to also play it safe, so the difference in the agents' progress remains constant. Doing so guarantees that the leader wins, so the follower's continuation value is strictly negative, contradicting the assumption that the follower chooses to stay!

The second observation characterizes the leader's risk attitude: he is strictly risk averse whenever the follower chooses to stay in the contest.

Lemma 2. *In any well-behaved MPE, the leader's value function is strictly concave. Formally, $V_i''(\Delta k) < 0$ wherever V_i'' exists on $(0, k_j^*)$.*

The intuition behind Lemma 2 is as follows. Suppose that the leader's value function is weakly convex over a range. We know from Lemma 1 that the follower takes the risky move, and one response available to the leader is to also take the risky move. Doing so ensures that the leader is not affected by the mean term, while the variance term helps the leader ($\sigma^2 V_i''' \geq 0$). It follows that the leader's continuation value is at least as good as the value of winning the contest for sure, again contradicting the assumption that the follower chooses to stay.

Combining the above two observations, we greatly simplify the agents' strategies in any well-behaved MPE. Specifically, Lemma 1 shows that the follower always takes the risky move, while Lemma 2 shows that

the leader's behavior switches at most once. Therefore, any well-behaved MPE is completely specified by a pair of stopping boundaries (k_i^*, k_j^*) , and possibly a pair of switching points (k_i^{**}, k_j^{**}) . By further deriving the best response functions for $(k_i^*, k_i^{**}, k_j^*, k_j^{**})$ from the boundary conditions of value matching and smooth pasting, we show that there exists a unique well-behaved MPE in each of the three cases, and it is symmetric. This strengthens Propositions 1, 2 and 3, respectively, into the following forms.

Proposition 1' (Low return case). *If the risky move has a low return, then there exists a unique well-behaved MPE, and it is symmetric and given by the following strategy profile:*

Drop out iff $\Delta k \leq -k^$, and otherwise $s_i^*(\Delta k) = 1$ iff $\Delta k < 0$, for some $k^* > 0$.*

Proposition 2' (High return case). *If the risky move has a high return, then there exists a unique well-behaved MPE, and it is symmetric and given by the following strategy profile:*

Drop out iff $\Delta k \leq -k^$, and otherwise $s_i^*(\Delta k) = 1$, for some $k^* > 0$.*

Proposition 3' (Medium return case). *If the risky move has a medium return, then there exists a unique well-behaved MPE, and it is symmetric and given by the following strategy profile:*

Drop out iff $\Delta k \leq -k^$, and otherwise $s_i^*(\Delta k) = 1$ iff $(\Delta k < 0 \text{ or } \Delta k \geq k^{**})$, for some $k^* > k^{**} > 0$.*

We have shown that the contest game has a unique well-behaved MPE, with equilibrium strategies as characterized in the previous subsections. We are currently working on a natural follow-up question: Are all MPEs well-behaved? In other words, does there exist an MPE that does not satisfy the HJB equations? While we cannot provide a concrete answer right now, we conclude this section with some conjectures and possible methods for verifying them.

Harris (1993) develops a theory of stochastic differential games in one dimension in which players' actions affect the state in a very general way. The paper proves that all MPEs satisfy the HJB equations (Harris, 1993, Theorems 5.6 and 6.5), as long as the regularity conditions there are satisfied. Our baseline model largely fits this general framework, with only one exception: if both agents play it safe, the system has no variance. This makes the analysis delicate because the order of the associated ODEs can change.

In subsequent research, we hope to prove results analogous to Harris (1993, Theorems 5.6 and 6.5), to show that all MPEs in our contest model are well-behaved.

4 Optimal Prize Allocation

In the previous section, we have focused on a stylized model and identified two motivations for taking the risky move: the Hail Mary motive and the preemptive motive. This section studies how that behavior shapes the optimal prize allocation, through an extended version of the baseline model to which all of the previous section's derivations and intuitions carry over. The main result is Proposition 5, which shows that “winner-takes-all” is optimal only when the designer's budget is small.

4.1 Extending Baseline Model

We extend the baseline model in two ways. First, the loser may also receive a prize, so the winner gets $\bar{P}$ and the loser gets $\underline{P}$, with $\bar{P} \geq \underline{P} \geq 0$. Second, the follower may also succeed, so each agent's hazard rate is $\bar{\lambda}$ when leading and $\underline{\lambda}$ when following, with $\bar{\lambda} > \underline{\lambda} \geq 0$ and the two agents equally likely to lead at a tie.

The derivations and intuitions in Section 3 are adapted by replacing the profitability of the contest $\phi = \lambda P/c$ with its generalized version

$$\bar{\phi} \equiv \frac{(\bar{\lambda} - \underline{\lambda})(\bar{P} - \underline{P})}{c - (\underline{\lambda}\bar{P} + \bar{\lambda}\underline{P})}. \quad (2)$$

and by replacing the effective discount rate $r + \lambda$ with $r + \bar{\lambda} + \underline{\lambda}$ throughout. The definition of the three cases. The definition of the three cases (Definition 1) is adapted accordingly.

Definition 1' (Low/medium/high return).

1. *The risky move has a low return if*

$$\pi \leq 0.$$

2. *The risky move has a medium return if¹*

$$0 < \frac{\pi}{\sigma} < \frac{\sqrt{r + \bar{\lambda} + \underline{\lambda}}}{2} \cdot f(\bar{\phi}).$$

¹Recall that the closed-form expression of $f(\cdot)$ is as follows:

$$f(\bar{\phi}) = \frac{\sqrt{2\bar{\phi}(\sqrt{\bar{\phi}^2 + 8 + \bar{\phi}} - 8)}}{\sqrt{\bar{\phi}^2 + 8 + \bar{\phi}}}.$$

3. *The risky move has a high return if*

$$\frac{\pi}{\sigma} \geq \frac{\sqrt{r + \bar{\lambda} + \underline{\lambda}}}{2} \cdot f(\bar{\phi}).$$

With this adapted definition, the three types of MPE carry over from the previous section unchanged, each parametrized by k^* and, in the medium return case, also by k^{**} .

4.2 A Contest Design Exercise

Suppose that the designer can choose any pair of prizes $(\bar{P}, \underline{P})$ subject to a fixed budget, $\bar{P} + \underline{P} \leq B$, and assume the budget satisfies $\bar{\lambda}B > c > \underline{\lambda}B$. This assumption rules out two degenerate cases. If $\bar{\lambda}B < c$, then even the largest feasible prize to the leader fails to cover his cost, so both agents drop out immediately. If $\underline{\lambda}B > c$, then both agents find it worthwhile to stay in the contest forever however the prize is split, so the design problem is trivial.

We consider three objectives a designer might reasonably pursue:

1. Minimize the expected time needed to achieve success (as in an innovation race).
2. Maximize the expected time during which both agents stay in the contest.
3. Maximize the continuation region (parametrized by k^*).

In our contest model, these objectives turn out to coincide.

Proposition 4 (Equivalence of objectives). *The designer's three objectives are equivalent:*

1. *Minimize the expected time needed to achieve success.*
2. *Maximize the expected time during which both agents stay in the contest.*
3. *Maximize the continuation region.*

The proof relies on a single monotonicity property. Let $\tau_{k^*} \equiv \inf \{t \geq 0 : |k_i(t) - k_j(t)| \geq k^*\}$ denote the first hitting time to the stopping boundary. In each of the three types of equilibria, $k_i - k_j$ has almost surely continuous sample paths, so τ_{k^*} is strictly increasing in k^* in the sense of first-order stochastic dominance. A larger continuation region therefore delays the follower's exit, which establishes the equivalence of Objectives 2 and 3. It also accelerates success: success arrives at rate $(\bar{\lambda} + \underline{\lambda})$ while both agents remain in the

contest and at the slower rate $\bar{\lambda}$ once the follower exits, so a larger k^* shortens the expected time to success. This establishes the equivalence of Objectives 1 and 3, and hence of all three.

We can now solve for the designer’s optimal prize allocation. Proposition 5 shows that “winner-takes-all” is optimal only when the designer’s budget is small.

Proposition 5 (Optimal prize allocation).

1. If $(\bar{\lambda} + \underline{\lambda})B/2 < c$, then it is optimal to set $\bar{P} = B$ and $\underline{P} = 0$.
2. If $(\bar{\lambda} + \underline{\lambda})B/2 > c$, then it is optimal to set $\bar{P} = B/2 + \varepsilon$ and $\underline{P} = B/2 - \varepsilon$, with sufficiently small $\varepsilon > 0$.

The intuition is as follows. When the budget is small, the designer cannot afford to keep both agents in the contest forever, and so chooses $(\bar{P}, \underline{P})$ to maximize the continuation region k^* . Since k^* is strictly increasing in the profitability of contest $\bar{\phi}$ defined in equation (2), this maximization yields $\bar{P} = B$ and $\underline{P} = 0$, that is, “winner-takes-all”. Such an allocation always induces a finite stopping boundary, however, and becomes suboptimal once the budget is large enough that some split keeps both agents in the contest forever.

5 Conclusion

We study a dynamic contest in which two agents compete to be the first to reach a goal, each choosing between a safe action that delivers deterministic progress and a risky move that stochastically shifts their relative position. We characterize three types of equilibria and identify two motives for the risky move: the follower’s Hail Mary motive and the leader’s preemptive motive. Turning to prize design, we find that “winner-takes-all” is optimal only when the designer’s budget is small.

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Appendix A Proofs of Results in the Main Text

A.1 Proofs for Section 3

Suppose that agent j plays according to some Markov strategy $(\hat{k}_j, s_j)$. Consider agent i 's best response within the class of Markov strategies, and let $\hat{k}_i$ denote agent i 's candidate stopping boundary. In the continuation region $-\hat{k}_i < \Delta k < \hat{k}_j$ and away from the tie, agent i 's value function $V_i(\Delta k)$ satisfies the following HJB equation:

$$rV_i dt = \begin{cases} \max_{a_i} \{[\lambda(P - V_i) - c] dt + \mathbb{E}[dV_i]\}, & 0 < \Delta k < \hat{k}_j, \\ \max_{a_i} \{[\lambda(0 - V_i) - c] dt + \mathbb{E}[dV_i]\}, & -\hat{k}_i < \Delta k < 0. \end{cases}$$

Since $\Delta k = k_i - k_j$ and $dk_i = (1 + a_i\pi) dt + a_i\sigma dw_i$, Itô's lemma implies that

$$\begin{aligned} dV_i &= V'_i dk_i - V'_i dk_j + \left(\frac{1}{2} a_i^2 \sigma^2 V''_i + \frac{1}{2} a_j^2 \sigma^2 V''_i \right) dt \\ \implies \mathbb{E}[dV_i] &= \left[(a_i - a_j) \pi V'_i + \frac{1}{2} (a_i^2 + a_j^2) \sigma^2 V''_i \right] dt. \end{aligned}$$

For notational simplicity, write s_j for $s_j(-\Delta k)$. We can then rewrite the HJB equation as

$$rV_i = \begin{cases} \max_{a_i} \left\{ \lambda(P - V_i) - c + (a_i - s_j) \pi V'_i + \frac{1}{2} (a_i^2 + s_j^2) \sigma^2 V''_i \right\}, & 0 < \Delta k < \hat{k}_j, \\ \max_{a_i} \left\{ \lambda(0 - V_i) - c + (a_i - s_j) \pi V'_i + \frac{1}{2} (a_i^2 + s_j^2) \sigma^2 V''_i \right\}, & -\hat{k}_i < \Delta k < 0, \end{cases} \quad (\text{A.1})$$

which combines the expected current flow payoff with the expected change in the continuation value due to the drift and diffusion of Δk . Thus any best-response action rule $s_i^*(\cdot)$ satisfies

$$s_i^* = \begin{cases} 1, & \pi V'_i + \frac{1}{2} \sigma^2 V''_i > 0, \\ \text{indifferent}, & \pi V'_i + \frac{1}{2} \sigma^2 V''_i = 0, \\ 0, & \pi V'_i + \frac{1}{2} \sigma^2 V''_i < 0. \end{cases} \quad (\text{A.2})$$

The value function solves (A.1) subject to value matching at the upper boundary, where agent j drops out:

$$V_i(\hat{k}_j) = \frac{\lambda P - c}{r + \lambda}. \quad (\text{A.3})$$

It also satisfies value matching and smooth pasting at agent i 's optimal dropout boundary k_i^* :

$$V_i(-k_i^*) = V'_i(-k_i^*) = 0. \quad (\text{A.4})$$

The upper-boundary value follows because, once agent j drops out, agent i continues alone, pays flow cost c , and succeeds at rate λ . Thus, by equation (1), the value of winning the contest for sure is

$$\bar{V} \equiv \mathbb{E}_{\tau \sim \text{Exp}(\lambda)} \left[e^{-r\tau} P - \frac{c}{r} (1 - e^{-r\tau}) \right] = \int_0^\infty \left(e^{-r\tau} P - \frac{c}{r} (1 - e^{-r\tau}) \right) \lambda e^{-\lambda\tau} d\tau = \frac{\lambda P - c}{r + \lambda}. \quad (\text{A.5})$$

The lower-boundary value is zero because dropping out immediately yields zero, and smooth pasting is the corresponding optimal-stopping condition.

Equations (A.3) and (A.4) pin down the values at the two endpoints of the continuation region. It remains to paste the one-sided solutions together at the tie $\Delta k = 0$. Under the fair tie-breaking convention, the expected value at a tie is $V_i(0) = \frac{1}{2} \lim_{\Delta k \downarrow 0} V_i(\Delta k) + \frac{1}{2} \lim_{\Delta k \uparrow 0} V_i(\Delta k)$. Moreover, in each candidate profile below, at least one agent takes the risky move on each side of the tie, so the diffusion is strictly positive in a deleted neighborhood of the tie. Hence V_i is continuously differentiable across the tie:

$$\lim_{\Delta k \downarrow 0} V_i(\Delta k) = \lim_{\Delta k \uparrow 0} V_i(\Delta k), \quad \lim_{\Delta k \downarrow 0} V'_i(\Delta k) = \lim_{\Delta k \uparrow 0} V'_i(\Delta k). \quad (\text{A.6})$$

Proof of Proposition 1. Consider the candidate equilibrium strategy:

$$\text{Drop out iff } \Delta k \leq -k^*, \text{ and otherwise } s_i^*(\Delta k) = 1 \text{ iff } \Delta k < 0.$$

Under this strategy, the HJB equation (A.1) becomes

$$rV_i = \begin{cases} \lambda(P - V_i) - c - \pi V'_i + \frac{1}{2} \sigma^2 V''_i, & 0 < \Delta k < k^*, \\ \lambda(0 - V_i) - c + \pi V'_i + \frac{1}{2} \sigma^2 V''_i, & -k^* < \Delta k < 0. \end{cases}$$

Solving these second-order linear ODEs subject to the boundary conditions (A.3), (A.4), and (A.6), we obtain

$$V_i(\Delta k) = \begin{cases} \frac{\lambda P - c}{r + \lambda} + C_1 e^{-\xi_+ \Delta k} + C_2 e^{-\xi_- \Delta k}, & 0 < \Delta k < k^*, \\ -\frac{c}{r + \lambda} + C_3 e^{\xi_+ \Delta k} + C_4 e^{\xi_- \Delta k}, & -k^* < \Delta k < 0, \end{cases}$$

where $\xi_+ > 0$ and $\xi_- < 0$ are the two roots of the characteristic equation

$$r = -\lambda + \pi \xi + \frac{1}{2} \sigma^2 \xi^2 \implies \xi_{\pm} = \frac{-\pi \pm \sqrt{\pi^2 + 2(r + \lambda)\sigma^2}}{\sigma^2}. \quad (\text{A.7})$$

The boundary conditions pin down C_1, C_2, C_3, C_4 and k^* ; eliminating the constants, $k^* > 0$ is the unique positive solution to

$$(\xi_+ + \xi_-)^2 e^{(\xi_+ + \xi_-)k^*} - 2\xi_+ \xi_- (e^{2\xi_+ k^*} + e^{2\xi_- k^*}) = \phi(\xi_+ - \xi_-) (\xi_+ e^{\xi_+ k^*} - \xi_- e^{\xi_- k^*}). \quad (\text{A.8})$$

We first show that the candidate value function is strictly increasing on each side of the tie. On the follower side $-k^* < \Delta k < 0$, the lower-boundary conditions (A.4) give

$$V'_i(\Delta k) = \xi_+ C_3 e^{\xi_+ \Delta k} + \xi_- C_4 e^{\xi_- \Delta k} = \frac{c}{r + \lambda} \cdot \frac{\xi_+ \xi_-}{\xi_+ - \xi_-} (e^{\xi_- (\Delta k + k^*)} - e^{\xi_+ (\Delta k + k^*)}) > 0,$$

since $\xi_+ \xi_- / (\xi_+ - \xi_-) < 0$ (as $\xi_+ > 0$ and $\xi_- < 0$) and $e^{\xi_- (\Delta k + k^*)} - e^{\xi_+ (\Delta k + k^*)} < 0$ (as $\Delta k + k^* > 0$). Hence $\lim_{\Delta k \uparrow 0} V'_i(\Delta k) > 0$, and by continuity of V'_i across the tie (A.6), $\lim_{\Delta k \downarrow 0} V'_i(\Delta k) > 0$ as well. On the leader side $0 < \Delta k < k^*$, the upper-boundary condition (A.3) gives $C_2 = -C_1 e^{-(\xi_+ - \xi_-)k^*}$, so

$$V'_i(\Delta k) = -\xi_+ C_1 e^{-\xi_+ \Delta k} - \xi_- C_2 e^{-\xi_- \Delta k} = C_1 (-\xi_+ e^{-\xi_+ \Delta k} + \xi_- e^{-(\xi_+ - \xi_-)k^*} e^{-\xi_- \Delta k}).$$

The bracketed factor is negative on $0 < \Delta k < k^*$, since both terms are. Letting $\Delta k \downarrow 0$ then gives $C_1 < 0$, since $\lim_{\Delta k \downarrow 0} V'_i(\Delta k) > 0$ and the bracket is negative. Hence $V'_i > 0$ throughout the leader side, so V_i is

strictly increasing on $(-k^*, 0) \cup (0, k^*)$. Since $V_i(-k^*) = 0$, it follows that $V_i(\Delta k) > 0$ for all $-k^* < \Delta k < k^*$, so agent i strictly prefers to stay at every interior state and is indifferent at $\Delta k = -k^*$. The stopping rule is therefore optimal, with $-k^*$ the optimal dropout boundary.

It remains to verify that neither agent wishes to deviate from the candidate action rule.

1. When leading ($0 < \Delta k < k^*$), agent i prefers $a_i = 0$ to $a_i = 1$, i.e., $\pi V'_i + \frac{1}{2}\sigma^2 V''_i \leq 0$.

Since $\pi \leq 0$ by assumption and $V'_i > 0$ by the monotonicity above, we have $\pi V'_i \leq 0$. It remains to show that $V''_i \leq 0$, i.e., that V_i is concave on the leader side. Using $C_2 = -C_1 e^{-(\xi_+ - \xi_-)k^*}$ from (A.3), the leader-side solution gives

$$V''_i(\Delta k) = \xi_+^2 C_1 e^{-\xi_+ \Delta k} + \xi_-^2 C_2 e^{-\xi_- \Delta k} = C_1 e^{-\xi_+ \Delta k} \left(\xi_+^2 - \xi_-^2 e^{-(\xi_+ - \xi_-)(k^* - \Delta k)} \right).$$

On $0 < \Delta k < k^*$ the exponent $-(\xi_+ - \xi_-)(k^* - \Delta k)$ is negative, so $e^{-(\xi_+ - \xi_-)(k^* - \Delta k)} \in (0, 1)$ and the bracketed factor is at least $\xi_+^2 - \xi_-^2$. Since $\xi_+ + \xi_- = -2\pi/\sigma^2 \geq 0$ in the low return case, $\xi_+^2 \geq \xi_-^2$, so the bracket is positive. Combined with $C_1 < 0$ from above, this yields $V''_i < 0$. Therefore $\pi V'_i + \frac{1}{2}\sigma^2 V''_i \leq 0$ throughout the leader side.

2. When following ($-k^* < \Delta k < 0$), agent i prefers $a_i = 1$ to $a_i = 0$, i.e., $\pi V'_i + \frac{1}{2}\sigma^2 V''_i \geq 0$.

Here the HJB equation reads $rV_i = \lambda(0 - V_i) - c + \pi V'_i + \frac{1}{2}\sigma^2 V''_i$, so $\pi V'_i + \frac{1}{2}\sigma^2 V''_i = (r + \lambda)V_i + c$. Since V_i is strictly increasing on the follower side with $V_i(-k^*) = 0$, we have $V_i(\Delta k) \geq 0$ there. Together with $c \geq 0$, this makes the right-hand side nonnegative, so $\pi V'_i + \frac{1}{2}\sigma^2 V''_i \geq 0$ throughout the follower side.

This completes the proof. $\square$

The comparative statics stated after Proposition 1 can be derived from the first-order condition (A.8).

Proof of Proposition 2. Consider the candidate equilibrium strategy:

$$\text{Drop out iff } \Delta k \leq -k^*, \text{ and otherwise } s_i^*(\Delta k) = 1.$$

Under this strategy, the HJB equation (A.1) becomes

$$rV_i = \begin{cases} \lambda(P - V_i) - c + \sigma^2 V''_i, & 0 < \Delta k < k^*, \\ \lambda(0 - V_i) - c + \sigma^2 V''_i, & -k^* < \Delta k < 0. \end{cases}$$

Solving these second-order linear ODEs subject to the boundary conditions (A.3), (A.4), and (A.6), we obtain

$$V_i(\Delta k) = \begin{cases} \frac{\lambda P - c}{r + \lambda} + C_1 e^{-\eta \Delta k} + C_2 e^{\eta \Delta k}, & 0 < \Delta k < k^*, \\ -\frac{c}{r + \lambda} + C_3 e^{\eta \Delta k} + C_4 e^{-\eta \Delta k}, & -k^* < \Delta k < 0, \end{cases}$$

where $\eta > 0$ is the positive root of the characteristic equation

$$r = -\lambda + \sigma^2 \eta^2 \quad \implies \quad \eta = \frac{\sqrt{r + \lambda}}{\sigma}. \quad (\text{A.9})$$

The boundary conditions pin down C_1, C_2, C_3, C_4 and k^* ; eliminating the constants, $k^* > 0$ is the unique

positive solution to¹²

$$e^{2\eta k^*} + e^{-2\eta k^*} = \phi \left(e^{\eta k^*} + e^{-\eta k^*} \right). \quad (\text{A.10})$$

As in the low return case, the candidate value function is strictly increasing on each side of the tie. On the follower side $-k^* < \Delta k < 0$, the lower-boundary conditions (A.4) give

$$V'_i(\Delta k) = \eta C_3 e^{\eta \Delta k} - \eta C_4 e^{-\eta \Delta k} = \frac{c \eta}{2(r + \lambda)} \left(e^{\eta(\Delta k + k^*)} - e^{-\eta(\Delta k + k^*)} \right) > 0,$$

since $\Delta k + k^* > 0$. Hence $\lim_{\Delta k \uparrow 0} V'_i(\Delta k) > 0$, and by continuity of V'_i across the tie (A.6), $\lim_{\Delta k \downarrow 0} V'_i(\Delta k) > 0$ as well. On the leader side $0 < \Delta k < k^*$, the upper-boundary condition (A.3) gives $C_1 = -C_2 e^{2\eta k^*}$, so

$$V'_i(\Delta k) = -\eta C_1 e^{-\eta \Delta k} + \eta C_2 e^{\eta \Delta k} = \eta C_2 \left(e^{2\eta k^*} e^{-\eta \Delta k} + e^{\eta \Delta k} \right).$$

Letting $\Delta k \downarrow 0$ then gives $C_2 > 0$, since $\lim_{\Delta k \downarrow 0} V'_i(\Delta k) > 0$ and the bracketed factor is positive. Hence $V'_i > 0$ throughout the leader side, so V_i is strictly increasing on $(-k^*, 0) \cup (0, k^*)$. Since $V_i(-k^*) = 0$, it follows that $V_i(\Delta k) > 0$ for all $-k^* < \Delta k < k^*$, so agent i strictly prefers to stay at every interior state and is indifferent at $\Delta k = -k^*$. The stopping rule is therefore optimal, with $-k^*$ the optimal dropout boundary.

It remains to verify that neither agent wishes to deviate from the candidate action rule. Define $G_i \equiv \pi V'_i + \frac{1}{2} \sigma^2 V''_i$, agent i 's marginal gain from taking the risky move. By (A.2), the candidate action rule is a best response iff $G_i \geq 0$ throughout the continuation region. On each side of the tie V_i solves a constant-coefficient ODE with characteristic roots $\pm \eta$, so G_i is a linear combination of $e^{\pm \eta \Delta k}$ and satisfies $G''_i = \eta^2 G_i$. Consequently G_i admits no negative interior minimum on either side: at any interior minimizer $G''_i \geq 0$, hence $G_i = G''_i / \eta^2 \geq 0$ there. It therefore suffices to verify $G_i \geq 0$ at the endpoints of each side.

On the leader side, value matching (A.3) and the leader-side HJB give

$$G_i(k^*) = \pi V'_i(k^*) + \frac{1}{2} ((r + \lambda) V_i(k^*) - (\lambda P - c)) = \pi V'_i(k^*) \geq 0,$$

since $\pi \geq 0$ and $V'_i(k^*) > 0$. At the tie, the leader-side solution gives

$$\lim_{\Delta k \downarrow 0} G_i(\Delta k) = \frac{c}{2(r + \lambda)} \left(e^{\eta k^*} - e^{-\eta k^*} \right) \left(\pi \eta - \frac{1}{2} (r + \lambda) \frac{e^{\eta k^*} - e^{-\eta k^*}}{e^{\eta k^*} + e^{-\eta k^*}} \right).$$

The factor $\frac{c}{2(r + \lambda)} (e^{\eta k^*} - e^{-\eta k^*})$ is positive, and by (A.10) the ratio $\frac{e^{\eta k^*} - e^{-\eta k^*}}{e^{\eta k^*} + e^{-\eta k^*}}$ equals $f(\phi)$, so

$$\lim_{\Delta k \downarrow 0} G_i(\Delta k) \geq 0 \quad \Longleftrightarrow \quad \frac{\pi}{\sigma} \geq \frac{\sqrt{r + \lambda}}{2} f(\phi),$$

which holds in the high return case. Hence $G_i \geq 0$ on the leader side.

On the follower side, value matching and smooth pasting (A.4) give $V_i(-k^*) = V'_i(-k^*) = 0$, so the follower-side HJB yields

$$G_i(-k^*) = \pi V'_i(-k^*) + \frac{1}{2} ((r + \lambda) V_i(-k^*) + c) = \frac{c}{2} \geq 0.$$

¹²In closed form,

$$k^* = \frac{1}{\eta} \log \frac{\sqrt{\phi^2 + 8} + \phi + \sqrt{2\phi(\sqrt{\phi^2 + 8} + \phi) - 8}}{4}.$$

At the tie, V'_i is continuous by (A.6), so the follower- and leader-side HJBs give

$$\lim_{\Delta k \uparrow 0} G_i(\Delta k) = \pi V'_i(0) + \frac{1}{2}[(r + \lambda)V_i(0) + c] = \lim_{\Delta k \downarrow 0} G_i(\Delta k) + \frac{\lambda P}{2} > 0,$$

where the inequality holds because $\lim_{\Delta k \downarrow 0} G_i(\Delta k) \geq 0$ from the leader side. Hence $G_i \geq 0$ on the follower side.

This completes the proof. $\square$

Proof of Proposition 3. Consider the candidate equilibrium strategy:

$$\text{Drop out iff } \Delta k \leq -k^*, \text{ and otherwise } s_i^*(\Delta k) = 1 \text{ iff } (\Delta k < 0 \text{ or } \Delta k \geq k^{**}).$$

Under this strategy, the HJB equation (A.1) becomes

$$rV_i = \begin{cases} \lambda(P - V_i) - c + \sigma^2 V_i'', & k^{**} < \Delta k < k^*, \\ \lambda(P - V_i) - c - \pi V'_i + \frac{1}{2}\sigma^2 V_i'', & 0 < \Delta k < k^{**}, \\ \lambda(0 - V_i) - c + \pi V'_i + \frac{1}{2}\sigma^2 V_i'', & -k^{**} < \Delta k < 0, \\ \lambda(0 - V_i) - c + \sigma^2 V_i'', & -k^* < \Delta k < -k^{**}. \end{cases}$$

Solving these second-order linear ODEs subject to value matching (A.3) at k^* , value matching and smooth pasting (A.4) at $-k^*$, continuity of V_i and V'_i at the tie (A.6) and at the switching thresholds $\pm k^{**}$, and the leader's indifference condition at the switch,

$$\pi V'_i(k^{**}) + \frac{1}{2}\sigma^2 V''_i(k^{**}) = 0, \quad (\text{A.11})$$

we obtain

$$V_i(\Delta k) = \begin{cases} \frac{\lambda P - c}{r + \lambda} + C_1 e^{-\eta \Delta k} + C_2 e^{\eta \Delta k}, & k^{**} < \Delta k < k^*, \\ \frac{\lambda P - c}{r + \lambda} + C_3 e^{-\xi_+ \Delta k} + C_4 e^{-\xi_- \Delta k}, & 0 < \Delta k < k^{**}, \\ -\frac{c}{r + \lambda} + C_5 e^{\xi_+ \Delta k} + C_6 e^{\xi_- \Delta k}, & -k^{**} < \Delta k < 0, \\ -\frac{c}{r + \lambda} + C_7 e^{\eta \Delta k} + C_8 e^{-\eta \Delta k}, & -k^* < \Delta k < -k^{**}, \end{cases}$$

where $\xi_+ > 0$, $\xi_- < 0$, and $\eta > 0$ are the roots from (A.7) and (A.9). These conditions pin down the eight constants together with k^* and k^{**} . In particular, the indifference condition (A.11) determines k^{**} through

$$\frac{e^{\eta(k^* - k^{**})} - e^{-\eta(k^* - k^{**})}}{e^{\eta(k^* - k^{**})} + e^{-\eta(k^* - k^{**})}} = \frac{\pi}{\sigma} \cdot \frac{2}{\sqrt{r + \lambda}} \implies k^{**} = k^* - \frac{1}{2\eta} \log \frac{1 + \frac{\pi}{\sigma} \cdot \frac{2}{\sqrt{r + \lambda}}}{1 - \frac{\pi}{\sigma} \cdot \frac{2}{\sqrt{r + \lambda}}}, \quad (\text{A.12})$$

which satisfies $k^{**} < k^*$ since $0 < \frac{\pi}{\sigma} \cdot \frac{2}{\sqrt{r + \lambda}} < 1$ in the medium return case.

As in the earlier cases, the candidate value function is strictly increasing on the continuation region. To see this, first consider the two outer regions. On the follower side, $-k^* < \Delta k < -k^{**}$, the ODE and the lower-boundary conditions (A.4) coincide with those of the follower side in the high return case; since the

two conditions at $-k^*$ alone determine the solution on this region, the same computation gives

$$V_i'(\Delta k) = \frac{c\eta}{2(r+\lambda)} \left(e^{\eta(\Delta k + k^*)} - e^{-\eta(\Delta k + k^*)} \right) > 0.$$

On the leader side, $k^{**} < \Delta k < k^*$, the upper-boundary condition (A.3) gives $C_2 = -C_1 e^{-2\eta k^*}$, and hence

$$V_i(\Delta k) = \bar{V} + C_1 e^{-\eta \Delta k} \left(1 - e^{-2\eta(k^* - \Delta k)} \right),$$

where $\bar{V}$ is the value of winning the contest for sure. For $k^{**} < \Delta k < k^*$, agent j stays in the contest, and under the candidate strategy the follower always takes the risky move, so the state has nondegenerate diffusion throughout the continuation region. With positive probability, agent j therefore takes the lead before success occurs and wins the contest, so $V_i(\Delta k) < \bar{V}$ on this region. The bracketed factor is strictly positive, so $C_1 < 0$. Therefore

$$V_i'(\Delta k) = -\eta C_1 \left(e^{-\eta \Delta k} + e^{-2\eta k^*} e^{\eta \Delta k} \right) > 0, \quad k^{**} < \Delta k < k^*.$$

Next, consider the two inner regions. On $-k^{**} < \Delta k < 0$ and $0 < \Delta k < k^{**}$, differentiating the corresponding ODEs gives, respectively,

$$\frac{1}{2}\sigma^2 V_i'''' + \pi V_i''' - (r+\lambda) V_i' = 0 \quad \text{and} \quad \frac{1}{2}\sigma^2 V_i'''' - \pi V_i''' - (r+\lambda) V_i' = 0.$$

Thus V_i' has no nonpositive interior minimum on either inner region unless it vanishes identically there: at such a minimum, $V_i'' = 0$ and $V_i''' \geq 0$, while the displayed ODEs then give $\frac{1}{2}\sigma^2 V_i'''' = (r+\lambda) V_i' \leq 0$; hence $V_i' = V_i'' = 0$ at the minimum, and uniqueness of solutions to the displayed equation, a second-order linear ODE for V_i' , forces $V_i' \equiv 0$ on that region. The identically zero case is excluded: the positivity established on the two outer regions and continuity of V_i' at the switching thresholds give $V_i'(-k^{**}) > 0$ and $V_i'(k^{**}) > 0$, whereas $V_i' \equiv 0$ on an inner region would force $V_i' = 0$ at the adjacent switching threshold.

We claim $V_i'(0) > 0$. Suppose, to the contrary, that $V_i'(0) \leq 0$. The minimum of V_i' over $[-k^{**}, k^{**}]$ is then nonpositive; since V_i' is positive at $\pm k^{**}$ and has no nonpositive interior minimum within either inner region, this minimum can occur only at the tie, so $\lim_{\Delta k \uparrow 0} V_i''(\Delta k) \leq 0 \leq \lim_{\Delta k \downarrow 0} V_i''(\Delta k)$. On the other hand, subtracting the follower-side HJB from the leader-side HJB at the tie, using continuity of V_i and V_i' (A.6), gives

$$\frac{1}{2}\sigma^2 \left(\lim_{\Delta k \downarrow 0} V_i''(\Delta k) - \lim_{\Delta k \uparrow 0} V_i''(\Delta k) \right) = 2\pi V_i'(0) - \lambda P < 0,$$

where the inequality holds because $V_i'(0) \leq 0$ and $\pi > 0$. The left-hand side, however, is nonnegative, a contradiction. Therefore $V_i'(0) > 0$. Each inner region now has $V_i' > 0$ at both endpoints, so a nonpositive value in its interior would produce a nonpositive interior minimum, which we have excluded. Hence $V_i' > 0$ on $(-k^{**}, 0)$ and $(0, k^{**})$, and altogether $V_i' > 0$ on $(-k^*, k^*)$, so V_i is strictly increasing on the continuation region. Since $V_i(-k^*) = 0$, it follows that $V_i(\Delta k) > 0$ for all $-k^* < \Delta k < k^*$, so agent i strictly prefers to stay at every interior state and is indifferent at $\Delta k = -k^*$. The stopping rule is therefore optimal, with $-k^*$ the optimal dropout boundary.

It remains to verify that neither agent wishes to deviate from the candidate action rule. As in the proof of Proposition 2, let $G_i \equiv \pi V_i' + \frac{1}{2}\sigma^2 V_i''$ denote agent i 's marginal gain from taking the risky move. By (A.2), the candidate action rule is a best response iff $G_i \geq 0$ on $(-k^*, 0)$ and on $k^{**} < \Delta k < k^*$, where the candidate prescribes the risky move, and $G_i \leq 0$ on $0 < \Delta k < k^{**}$, where it prescribes the safe action.

On the follower side, the HJBs deliver the sign directly. On $-k^{**} < \Delta k < 0$, the HJB rearranges to $G_i = (r+\lambda) V_i + c > 0$, since $V_i > 0$. On $-k^* < \Delta k < -k^{**}$, the HJB gives $\sigma^2 V_i'' = (r+\lambda) V_i + c$, so

$G_i = \pi V'_i + \frac{1}{2}((r + \lambda) V_i + c) > 0$, where the inequality holds because $\pi > 0$ in the medium return case, $V'_i > 0$ by the monotonicity above, and $V_i > 0$. Hence $G_i > 0$ on the follower side.

We turn to the leader side, and first prove that G_i is continuous at the switching threshold k^{**} and equals zero there. Equating the two HJBs at k^{**} , using continuity of V_i and V'_i , gives

$$\sigma^2 \lim_{\Delta k \downarrow k^{**}} V''_i(\Delta k) = \frac{1}{2} \sigma^2 \lim_{\Delta k \uparrow k^{**}} V''_i(\Delta k) - \pi V'_i(k^{**}),$$

so the indifference condition (A.11) holds as $\Delta k \downarrow k^{**}$ iff it holds as $\Delta k \uparrow k^{**}$, in which case both one-sided limits of V''_i equal $-\frac{2\pi}{\sigma^2} V'_i(k^{**})$. Hence V''_i , and therefore G_i , is continuous at k^{**} , with $G_i(k^{**}) = 0$.

On $k^{**} < \Delta k < k^*$, both agents take the risky move, exactly as on the continuation region in the high return case. As in the proof of Proposition 2, V_i solves a constant-coefficient ODE with characteristic roots $\pm\eta$, so G_i is a linear combination of $e^{\pm\eta\Delta k}$ and satisfies $G''_i = \eta^2 G_i$; hence G_i has no negative interior minimum, and it suffices to verify $G_i \geq 0$ at the endpoints. At k^{**} , $G_i(k^{**}) = 0$. At k^* , value matching (A.3) and the HJB give

$$G_i(k^*) = \pi V'_i(k^*) + \frac{1}{2}((r + \lambda) V_i(k^*) - (\lambda P - c)) = \pi V'_i(k^*) > 0,$$

since $\pi > 0$ and $V'_i(k^*) > 0$. Hence $G_i \geq 0$ on $k^{**} < \Delta k < k^*$.

On $0 < \Delta k < k^{**}$, V_i is a constant plus a linear combination of $e^{-\xi_+\Delta k}$ and $e^{-\xi_-\Delta k}$, so G_i is a linear combination of $e^{-\xi_+\Delta k}$ and $e^{-\xi_-\Delta k}$. Since $G_i(k^{**}) = 0$, there is a constant A such that

$$G_i(\Delta k) = A \left(e^{-\xi_-(\Delta k - k^{**})} - e^{-\xi_+(\Delta k - k^{**})} \right).$$

For $0 < \Delta k < k^{**}$ we have $\Delta k - k^{**} < 0$, and since $\xi_+ > 0$ and $\xi_- < 0$ the bracket is strictly negative, so G_i has the sign of $-A$ on this region. It therefore suffices to show $A > 0$. Differentiating gives $\lim_{\Delta k \uparrow k^{**}} G'_i(\Delta k) = A(\xi_+ - \xi_-)$, so A has the sign of this limit. To compute it, differentiate the HJB on $0 < \Delta k < k^{**}$, which gives $\frac{1}{2}\sigma^2 V'''_i = (r + \lambda) V'_i + \pi V''_i$, and hence

$$G'_i = \pi V''_i + \frac{1}{2}\sigma^2 V'''_i = (r + \lambda) V'_i + 2\pi V''_i.$$

Evaluating at k^{**} and substituting $V''_i(k^{**}) = -\frac{2\pi}{\sigma^2} V'_i(k^{**})$ from (A.11) gives

$$\lim_{\Delta k \uparrow k^{**}} G'_i(\Delta k) = \left(r + \lambda - \frac{4\pi^2}{\sigma^2} \right) V'_i(k^{**}) > 0,$$

where the inequality holds because $V'_i(k^{**}) > 0$ by the monotonicity above and because the medium return case gives $\frac{\pi}{\sigma} < \frac{\sqrt{r+\lambda}}{2} f(\phi) < \frac{\sqrt{r+\lambda}}{2}$, using $f(\phi) < 1$. Hence $A > 0$, so $G_i < 0$ on $0 < \Delta k < k^{**}$. □

This completes the proof.

We demonstrate two additional properties of MPEs. The first lemma shows that the cutoff representation is without loss.

Lemma A.1 (Cutoff representation). *In any MPE, agent i 's dropout region is downward closed: if dropout is optimal at Δk , it is optimal at every $\Delta k' < \Delta k$. Under the convention that an agent drops out whenever indifferent, there exists a cutoff $k_i^* \in [0, \infty]$ such that agent i stays in the contest iff $\Delta k > -k_i^*$.*

Proof. Let S_i denote the set of states at which dropout is optimal for agent i .

We first show that no strictly positive state belongs to S_i . Fix $\Delta k > 0$, so agent i is the leader. Consider the strategy in which agent i active, takes the safe action, and drops out if the state reaches 0 before

success occurs. Let τ_0 be the first time, under this strategy, at which the state reaches 0. While $t < \tau_0$ the state is positive and agent i is the leader, so he succeeds at rate λ . Discounting at r and weighting by the probability $e^{-\lambda t}$ that success has not yet occurred, the agent's value is at least

$$\mathbb{E} \left[\int_0^{\tau_0} e^{-(r+\lambda)t} (\lambda P - c) dt \right].$$

Because the state has continuous sample paths and starts at $\Delta k > 0$, we have $\tau_0 > 0$ with positive probability; since $\lambda P > c$, this value is strictly positive. Hence dropout is not optimal at Δk , so $\Delta k \notin S_i$.

Now take any $\Delta k \leq 0$ at which dropout is optimal, and fix any lower state $\Delta k' < \Delta k$. We show dropout is also optimal at $\Delta k'$. From $\Delta k'$ agent i is the follower. Let $\tau_{\Delta k}$ be the first time the state reaches Δk and let τ be the success time. Before $\tau_{\Delta k}$ the state satisfies $\Delta k(t) < \Delta k \leq 0$, so agent i is never the leader and collects no prize. Hence, before $\tau_{\Delta k} \wedge \tau$, agent i only pays flow costs. If success occurs first, agent j wins and agent i receives nothing; if instead the state reaches Δk first, then by stationarity agent i 's optimal continuation value there is zero, since dropout is optimal at Δk . In either case the value of remaining active at $\Delta k'$ is at most zero. Since agent i can drop out immediately and obtain zero, dropout is optimal at $\Delta k'$. Hence S_i is downward closed.

Set $-k_i^* \equiv \sup S_i$. Downward closedness implies that dropout is optimal at every $\Delta k < -k_i^*$ and not optimal at every $\Delta k > -k_i^*$. Under the convention that the agent drops out whenever dropout is optimal, agent i stays in the contest iff $\Delta k > -k_i^*$. The first step of the proof shows that no strictly positive state belongs to S_i , so $-k_i^* \leq 0$, or equivalently $k_i^* \geq 0$.

This completes the proof. $\square$

The next lemma bounds the sum of the agents' values inside the continuation region.

Lemma A.2 (Value-sum bound). *In any MPE, at every interior state the two agents' values sum to strictly less than the value of winning the contest for sure: $V_i(\Delta k) + V_j(-\Delta k) < \bar{V}$ for every $-k_i^* < \Delta k < k_j^*$. In particular, whenever agent i is the leader and agent j stays in the contest, $V_i(\Delta k) < \bar{V}$ for every $0 < \Delta k < k_j^*$.*

Proof. Fix an interior state $-k_i^* < \Delta k < k_j^*$. Let τ_S be the success time and let τ_D be the first time either agent drops out. Under equilibrium play, exactly one agent wins the prize P . Until the first dropout or success, both agents pay the flow cost; after the first dropout, the remaining agent pays the flow cost until success. Therefore

$$V_i(\Delta k) + V_j(-\Delta k) = \mathbb{E} \left[e^{-r\tau_S} P - \int_0^{\tau_S} e^{-rt} c dt - \int_0^{\tau_D \wedge \tau_S} e^{-rt} c dt \right].$$

Success occurs at rate λ while at least one agent stays in the contest, and at least one does so until success, since after any dropout the sole survivor stays in and obtains $\bar{V} > 0$. Hence τ_S is exponential with rate λ , and

$$\mathbb{E} \left[e^{-r\tau_S} P - \int_0^{\tau_S} e^{-rt} c dt \right] = \frac{\lambda P - c}{r + \lambda} = \bar{V}.$$

For the last term, the state process starts strictly inside the continuation region and has continuous sample paths, while the success time has no atom at zero; hence $\tau_D \wedge \tau_S > 0$ almost surely. Since $c > 0$,

$$\mathbb{E} \left[\int_0^{\tau_D \wedge \tau_S} e^{-rt} c dt \right] > 0.$$

Therefore, $V_i(\Delta k) + V_j(-\Delta k) < \bar{V}$.

Now suppose $0 < \Delta k < k_j^*$. Since agent j can drop out at any time for a payoff of zero, $V_j(-\Delta k) \geq 0$. Combined with the bound just established, this gives $V_i(\Delta k) < \bar{V}$.

This completes the proof. $\square$

These two observations play a useful role in the proofs of Lemmas 1 and 2.

Proof of Lemma 1. Suppose, to the contrary, that $s_i^*(\Delta k) = 0$ for some $-k_i^* < \Delta k < 0$, where agent i is the follower. At the corresponding state $k_j - k_i = -\Delta k > 0$, agent j is the leader. If agent j also takes the safe action, both agents make deterministic progress at the same rate, so the state stays fixed at $-\Delta k$; agent j remains the leader and succeeds at rate λ until the contest ends. Hence agent j can guarantee the value of winning the contest for sure: $V_j(-\Delta k) \geq \bar{V}$. But agent j leads while agent i stays in the contest, so Lemma A.2 gives $V_j(-\Delta k) < \bar{V}$, a contradiction! $\square$

Proof of Lemma 2. Suppose, to the contrary, that $V_i''(\Delta k) \geq 0$ at some $\Delta k \in (0, k_j^*)$ where V_i'' exists and agent i is the leader. By Lemma 1 the follower takes the risky move, so taking the risky move is feasible for agent i and leaves his drift unaffected, while the variance term contributes $\sigma^2 V_i'' \geq 0$. Hence the HJB gives

$$rV_i(\Delta k) \geq \lambda(P - V_i(\Delta k)) - c + \sigma^2 V_i''(\Delta k) \geq \lambda(P - V_i(\Delta k)) - c,$$

so $V_i(\Delta k) \geq \bar{V}$. But agent i leads while agent j stays in the contest, so Lemma A.2 gives $V_i(\Delta k) < \bar{V}$, a contradiction! $\square$

The next lemma shows that each agent's value is strictly increasing on the continuation region.

Lemma A.3 (Monotonicity). *In any MPE, each agent's value function is strictly increasing on the continuation region: $V_i(x) < V_i(y)$ for all $-k_i^* < x < y < k_j^*$.*

Proof. Fix an MPE. First consider the follower side, and take two states $-k_i^* < x < y < 0$. We bound the value of any continuation from x by what it can collect on the way to y . Starting from x , let τ_y be the first time the state reaches y , let τ_S be the success time, and let τ_D^i be the time agent i drops out. Before τ_y agent i is the follower, so he wins no prize: if success comes first agent j wins, if agent i drops out first he gets zero, and if the state reaches y first his continuation payoff there is $V_i(y)$. He pays the flow cost throughout. Hence agent i 's payoff from any strategy starting from x is at most

$$\mathbb{E} \left[e^{-r\tau_y} \mathbf{1}(\tau_y < \tau_S \wedge \tau_D^i) V_i(y) - \int_0^{\tau_y \wedge \tau_S \wedge \tau_D^i} e^{-rt} c dt \right] \leq \mathbb{E} \left[e^{-r\tau_y} \mathbf{1}(\tau_y < \tau_S \wedge \tau_D^i) \right] V_i(y),$$

where the second inequality drops the nonnegative cost term and uses $V_i(y) > 0$, which holds because y is interior to the continuation region, where staying is strictly better than the zero from dropping out.

It remains to bound the discounted hitting probability. The state has drift $(a_i - a_j)\pi$, of absolute value at most $|\pi|$, and squared diffusion coefficient $(a_i^2 + a_j^2)\sigma^2 \leq 2\sigma^2$, both bounded uniformly over feasible actions, and y lies a fixed distance $y - x > 0$ above x . Hence reaching y cannot be made arbitrarily fast, and there exists $\rho(x, y) \in (0, 1)$, depending only on $y - x$ and the primitives,¹³ with $\mathbb{E} \left[e^{-r\tau_y} \mathbf{1}(\tau_y < \tau_S \wedge \tau_D^i) \right] \leq \rho(x, y)$.

¹³Let

$$\beta \equiv \frac{-|\pi| + \sqrt{\pi^2 + 4r\sigma^2}}{2\sigma^2} > 0,$$

the positive root of $\beta|\pi| + \beta^2\sigma^2 - r = 0$. Consider $M(t) \equiv e^{\beta(\Delta k(t) - x) - rt}$, which starts at $M(0) = 1$ and, at the moment the state reaches y , equals $e^{\beta(y-x) - r\tau_y}$. By Itô's lemma, the drift coefficient of $M(t)$ is $\beta(a_i - a_j)\pi + \frac{1}{2}\beta^2(a_i^2 + a_j^2)\sigma^2 - r \leq \beta|\pi| + \beta^2\sigma^2 - r = 0$, using $|a_i - a_j| \leq 1$ and $a_i^2 + a_j^2 \leq 2$. Hence $M(t)$ is a positive supermartingale. Optional stopping at $\tau_y \wedge n$ gives $1 = M(0) \geq \mathbb{E} \left[M(\tau_y \wedge n) \right] \geq e^{\beta(y-x)} \mathbb{E} \left[e^{-r\tau_y} \mathbf{1}(\tau_y \leq n) \right]$, because $\Delta k(\tau_y) = y$ on $\{\tau_y \leq n\}$ and $M(n) \geq 0$ on $\{\tau_y > n\}$. Letting $n \rightarrow \infty$, $\mathbb{E} \left[e^{-r\tau_y} \mathbf{1}(\tau_y < \infty) \right] \leq e^{-\beta(y-x)} \equiv \rho(x, y) < 1$.

Substituting into the bound above, every strategy from x is worth at most $\rho(x, y) V_i(y)$, a bound independent of the strategy; therefore, $V_i(x) \leq \rho(x, y) V_i(y) < V_i(y)$, as desired.

Next consider the leader side. Take $0 < x < y < k_j^*$. By Lemma A.2, since agent i leads while agent j stays in the contest at x , we have $V_i(x) < \bar{V}$. Starting from y , consider the following feasible strategy for agent i : take the safe action until the first time the state reaches x , agent j drops out, or success occurs. If the state reaches x before success or agent j 's dropout, agent i follows his equilibrium continuation strategy from x and obtains $V_i(x)$. If agent j drops out first, agent i obtains $\bar{V}$. While the state remains in (x, k_j^*) , agent i is the leader, pays the flow cost c , and succeeds at rate λ .

Let $\tau \equiv \inf \{t \geq 0 : \Delta k(t) \leq x \text{ or } \Delta k(t) \geq k_j^*\}$ under this feasible strategy, excluding the success time. The value of this feasible strategy is

$$\mathbb{E} \left[\int_0^\tau e^{-(r+\lambda)t} (\lambda P - c) dt + e^{-(r+\lambda)\tau} \left(\mathbf{1}(\Delta k(\tau) = x) V_i(x) + \mathbf{1}(\Delta k(\tau) = k_j^*) \bar{V} \right) \right].$$

Since $\bar{V} > V_i(x)$, replacing the terminal payoff $\bar{V}$ after agent j 's dropout by $V_i(x)$ only lowers this value. Hence the feasible strategy is worth at least

$$\mathbb{E} \left[\int_0^\tau e^{-(r+\lambda)t} (\lambda P - c) dt + e^{-(r+\lambda)\tau} V_i(x) \right].$$

Using $\bar{V} = \frac{\lambda P - c}{r + \lambda}$, this lower bound equals $\bar{V} - \mathbb{E} \left[e^{-(r+\lambda)\tau} \right] (\bar{V} - V_i(x))$. Because $x < y < k_j^*$ and the state has continuous sample paths, $\tau > 0$ almost surely. Therefore, $\mathbb{E} \left[e^{-(r+\lambda)\tau} \right] < 1$. Together with $\bar{V} > V_i(x)$, this implies $\bar{V} - \mathbb{E} \left[e^{-(r+\lambda)\tau} \right] (\bar{V} - V_i(x)) > V_i(x)$. Thus agent i has a feasible strategy from y that yields strictly more than $V_i(x)$. Since $V_i(y)$ is the value of the optimal continuation from y , we obtain $V_i(y) > V_i(x)$, as desired.

Finally, consider $x < 0 < y$. Since V_i is strictly increasing on each side, the one-sided limits $\lim_{\Delta k \uparrow 0} V_i(\Delta k)$ and $\lim_{\Delta k \downarrow 0} V_i(\Delta k)$ exist, and the follower- and leader-side comparisons give $V_i(x) < \lim_{\Delta k \uparrow 0} V_i(\Delta k)$ and $\lim_{\Delta k \downarrow 0} V_i(\Delta k) < V_i(y)$. By Lemma 1 the follower takes the risky move, so the state has diffusion at least $\sigma > 0$ in a neighborhood of the tie; hence from either side at distance ε the hitting time of the tie vanishes as $\varepsilon \downarrow 0$, and the expected payoff accumulated before reaching it vanishes. The value therefore cannot jump across the tie, so $\lim_{\Delta k \uparrow 0} V_i(\Delta k) = \lim_{\Delta k \downarrow 0} V_i(\Delta k)$, giving $V_i(x) < V_i(y)$. Combining the follower side, the leader side, and the crossing case proves that V_i is strictly increasing on the continuation region.

This completes the proof. $\square$

Together, these lemmas significantly restrict the structure of any well-behaved MPE: the follower takes the risky move throughout (Lemma 1), the leader's value is strictly concave (Lemma 2), and each agent's value is strictly increasing up to his stopping boundary (Lemma A.3). The uniqueness results below build on this structure, beginning with the low return case.

Proof of Proposition 1'. Suppose that the agents play according to some well-behaved MPE $(k_i^*, s_i^*, k_j^*, s_j^*)$.

We first characterize the action rule. By Lemma 1, the follower takes the risky move throughout his continuation region: $s_i^*(\Delta k) = 1$ for every $-k_i^* < \Delta k < 0$. Now consider a leader state $0 < \Delta k < k_j^*$. Since the MPE is well-behaved, the value function solves the HJB equations in the piecewise classical sense; in particular, on each open interval on which the HJB has fixed coefficients, V_i solves a constant-coefficient linear ODE and is therefore continuously differentiable there. By Lemma A.3, V_i is strictly increasing on the continuation region, so $V_i'(\Delta k) \geq 0$ wherever the derivative exists. By Lemma 2, $V_i''(\Delta k) < 0$ wherever V_i'' exists. In the low return case $\pi \leq 0$, so $\pi V_i'(\Delta k) \leq 0$, while $\frac{1}{2} \sigma^2 V_i''(\Delta k) < 0$; hence $\pi V_i'(\Delta k) + \frac{1}{2} \sigma^2 V_i''(\Delta k) < 0$. By the best-response condition (A.2), agent i strictly prefers the safe action whenever he is the leader.

Thus, except possibly at the points where V_i'' fails to exist, which carry no payoff consequence, $s_i^*(\Delta k) = 0$. Hence any well-behaved MPE in the low return case has the same action structure: each agent takes the risky move if and only if he is the follower.

Given agent j 's stopping boundary $k_j^* > 0$, agent i 's value function solves the second-order linear ODEs

$$rV_i = \begin{cases} \lambda(P - V_i) - c - \pi V_i' + \frac{1}{2}\sigma^2 V_i'', & 0 < \Delta k < k_j^*, \\ \lambda(0 - V_i) - c + \pi V_i' + \frac{1}{2}\sigma^2 V_i'', & -k_i^* < \Delta k < 0, \end{cases}$$

subject to the boundary and tie-pasting conditions (A.3), (A.4), and (A.6). Hence

$$V_i(\Delta k) = \begin{cases} \frac{\lambda P - c}{r + \lambda} + C_{1i}e^{-\xi_+ \Delta k} + C_{2i}e^{-\xi_- \Delta k}, & 0 < \Delta k < k_j^*, \\ -\frac{c}{r + \lambda} + C_{3i}e^{\xi_+ \Delta k} + C_{4i}e^{\xi_- \Delta k}, & -k_i^* < \Delta k < 0, \end{cases}$$

where $\xi_+ > 0$ and $\xi_- < 0$ are defined in (A.7). Eliminating the constants using the boundary and tie-pasting conditions gives the equation determining agent i 's best-response stopping boundary as a function of k_j^* :

$$(\xi_+ + \xi_-) \left(\xi_+ e^{\xi_- k_i^* + \xi_+ k_j^*} + \xi_- e^{\xi_+ k_i^* + \xi_- k_j^*} \right) - 2\xi_+ \xi_- \left(e^{\xi_+ (k_i^* + k_j^*)} + e^{\xi_- (k_i^* + k_j^*)} \right) = \phi(\xi_+ - \xi_-) \left(\xi_+ e^{\xi_+ k_j^*} - \xi_- e^{\xi_- k_j^*} \right), \quad (\text{A.13})$$

Denote the resulting best response by $k_i^* = BR(k_j^*)$.

We now verify the relevant properties of BR . Write $p \equiv \xi_+ > 0$ and $q \equiv -\xi_- > 0$. In the low return case $\pi \leq 0$, so $p - q = \xi_+ + \xi_- = -2\pi/\sigma^2 \geq 0$. Let $\Theta(k_i, k_j)$ denote the left-hand side of (A.13) minus the right-hand side, after substituting $\xi_+ = p$ and $\xi_- = -q$:

$$\Theta(k_i, k_j) = (p - q) \left(p e^{-qk_i + pk_j} - q e^{pk_i - qk_j} \right) + 2pq \left(e^{p(k_i + k_j)} + e^{-q(k_i + k_j)} \right) - \phi(p + q) \left(p e^{pk_j} + q e^{-qk_j} \right).$$

The best-response boundary is determined by $\Theta(k_i, k_j) = 0$. For fixed $k_j > 0$, we have $\Theta(0, k_j) < 0$ because $\phi > 1$, while $\Theta(k_i, k_j) \rightarrow \infty$ as $k_i \rightarrow \infty$. Hence a positive solution exists. We next show that it is unique and that the best response is decreasing. Set $X \equiv e^{(p+q)k_i} > 1$ and $Y \equiv e^{(p+q)k_j} > 1$. Differentiating with respect to k_i gives

$$\Theta_i = pqe^{-q(k_i + k_j)} (2pXY - (p - q)(X + Y) - 2q) = pqe^{-q(k_i + k_j)} (p(2XY - X - Y) + q(X + Y - 2)).$$

The bracket is strictly positive because $X, Y > 1$. Therefore $\Theta_i > 0$, so the positive solution is unique. By the implicit function theorem, BR is differentiable with $BR'(k_j) = -\Theta_j/\Theta_i$. Differentiating with respect to k_j and using $\Theta(k_i, k_j) = 0$ to substitute out ϕ ,

$$\Theta_j = \frac{pq(p + q)^2 (X - 1) e^{pk_j - qk_i}}{pY + q} > 0.$$

Moreover,

$$\Theta_i - \Theta_j = \frac{pq(Y - 1) e^{-q(k_i + k_j)}}{pY + q} \left(p^2 Y (2X - 1) + q(p - q)X + pqY + 2q^2 \right),$$

where the bracket is strictly positive using $X, Y > 1$ and $p - q \geq 0$. Hence $\Theta_i > \Theta_j > 0$, and therefore $BR'(k_j) = -\Theta_j/\Theta_i \in (-1, 0)$. Thus stopping boundaries are strategic substitutes, and the slope of the best

response is everywhere larger than -1 . Examining (A.13) as $k_j^* \downarrow 0$ and $k_j^* \rightarrow \infty$ gives $\lim_{k_j^* \downarrow 0} BR(k_j^*) \in (0, \infty)$ and $\lim_{k_j^* \rightarrow \infty} BR(k_j^*) \in (0, \infty)$.

In equilibrium, $k_i^* = BR(k_j^*)$ and $k_j^* = BR(k_i^*)$. Define $Z(k) \equiv BR(BR(k)) - k$. Since $BR' \in (-1, 0)$, we have $Z'(k) = BR'(BR(k))BR'(k) - 1 < 0$. The limiting properties of BR give $Z(0) > 0$ and $\lim_{k \rightarrow \infty} Z(k) < 0$. Therefore, there exists a unique $k^* \in (0, \infty)$ with $Z(k^*) = 0$. Moreover, if $k_i^* = BR(k_j^*)$ and $k_j^* = BR(k_i^*)$, then both k_i^* and k_j^* are zeros of Z . By uniqueness, $k_i^* = k_j^* = k^*$.

Substituting $k_i^* = k_j^* = k^*$ into (A.13) simplifies it to (A.8). Proposition 1 then verifies that the symmetric strategy profile with this stopping boundary is an MPE. Therefore, the low return case has a unique well-behaved MPE and it is symmetric.

This completes the proof. $\square$

The remaining two propositions, Propositions 2' and 3', rely on the following two lemmas.

Lemma A.4. Suppose $\frac{\pi}{\sigma} > \frac{\sqrt{r+\lambda}}{2}$. In any well-behaved MPE, the leader always takes the risky move. Formally, $s_i^*(\Delta k) = 1$ wherever V_i'' exists on $(0, k_j^*)$.

Proof. Define $G_i \equiv \pi V_i' + \frac{1}{2}\sigma^2 V_i''$, agent i 's marginal gain from taking the risky move. We prove the stronger claim, that $G_i(\Delta k) > 0$ wherever V_i'' exists on $(0, k_j^*)$, which by the best-response condition (A.2) gives $s_i^*(\Delta k) = 1$ there.

By Lemma 1, the follower always takes the risky move in his continuation region. Hence, by the HJB equation (A.1), on any open interval with fixed action a_i we have

$$rV_i = \lambda(P - V_i) - c + (a_i - 1)\pi V_i' + \frac{1}{2}(a_i^2 + 1)\sigma^2 V_i''.$$

Differentiating gives

$$(r + \lambda)V_i' = (a_i - 1)\pi V_i'' + \frac{1}{2}(a_i^2 + 1)\sigma^2 V_i'''.$$

Hence

$$G_i' = \pi V_i'' + \frac{1}{2}\sigma^2 V_i''' = \frac{(r + \lambda)V_i' + (a_i^2 - a_i + 2)\pi V_i''}{a_i^2 + 1}. \quad (\text{A.14})$$

Since $a_i \in \{0, 1\}$, $a_i^2 = a_i$ and the numerator is $(r + \lambda)V_i' + 2\pi V_i''$. Lemma A.3 gives $V_i' \geq 0$ and Lemma 2 gives $V_i'' < 0$ wherever these derivatives exist. At any such state where $G_i \leq 0$, dividing by $-V_i'' > 0$ and using $\frac{\pi}{\sigma} > \frac{\sqrt{r+\lambda}}{2}$ gives

$$\frac{V_i'}{-V_i''} \leq \frac{\frac{1}{2}\sigma^2}{\pi} < \frac{\pi}{\frac{1}{2}(r + \lambda)},$$

so $(r + \lambda)V_i' + 2\pi V_i'' < 0$ and hence

$$G_i' < 0 \quad \text{wherever} \quad G_i \leq 0. \quad (\text{A.15})$$

Fix a state $x \in (0, k_j^*)$ at which V_i'' exists, and suppose to the contrary that $G_i(x) \leq 0$. By (A.15), $G_i < 0$ at all higher states where V_i'' exists,¹⁴ so by the best-response condition (A.2) the leader takes the safe action

¹⁴At a state y where the leader switches actions, the boundary conditions give a jump in V_i'' such that $\lim_{\Delta k \downarrow y} G_i = \frac{1}{2} \lim_{\Delta k \uparrow y} G_i$ (safe to risky) or $\lim_{\Delta k \downarrow y} G_i = 2 \lim_{\Delta k \uparrow y} G_i$ (risky to safe). Both factors are positive, so a switch preserves the sign of G_i . Because optimality requires $G_i \leq 0$ on a safe side and $G_i \geq 0$ on a risky side, an actual switch between safe and risky can occur only when both one-sided limits vanish; see the proof of Proposition 3, where G_i is continuous at the switching state k^{**} with value zero.

throughout (x, k_j^*) . Value matching at the upper boundary (A.3) gives $V_i(k_j^*) = \bar{V}$. Setting $a_i = 0$ in the HJB equation (A.1) and evaluating this ODE as $\Delta k \uparrow k_j^*$ then gives

$$\lim_{\Delta k \uparrow k_j^*} \left(-\pi V_i' + \frac{1}{2} \sigma^2 V_i'' \right) = 0.$$

Since $\pi > 0$, $V_i' \geq 0$, and $V_i'' < 0$, each term inside the bracket is nonpositive, so their sum can converge to zero only if

$$\lim_{\Delta k \uparrow k_j^*} V_i' = \lim_{\Delta k \uparrow k_j^*} V_i'' = 0.$$

The only solution to the ODE with that limiting property is the constant solution $V_i(\Delta k) = \bar{V}$. But agent i leads while agent j stays in the contest, so Lemma A.2 gives $V_i(\Delta k) < \bar{V}$, a contradiction!

This completes the proof. $\square$

Lemma A.5. Suppose $0 < \frac{\pi}{\sigma} < \frac{\sqrt{r+\lambda}}{2}$. In any well-behaved MPE, the leader's risky region is upward closed: if the leader takes the risky move at some state, then he also takes it at all higher states. Formally, if $s_i^*(x) = 1$ for some $x \in (0, k_j^*)$ at which V_i'' exists, then $s_i^*(\Delta k) = 1$ wherever V_i'' exists on (x, k_j^*) .

Proof. Let $G_i \equiv \pi V_i' + \frac{1}{2} \sigma^2 V_i''$ as above, and suppose $s_i^*(x) = 1$ for some $x \in (0, k_j^*)$ at which V_i'' exists. The best-response condition (A.2) gives $G_i(x) \geq 0$. We show $G_i(\Delta k) > 0$ wherever V_i'' exists on (x, k_j^*) , which gives $s_i^*(\Delta k) = 1$ there.

By Lemma 1 the follower takes the risky move, so the identity (A.14) derived in the proof of Lemma A.4 holds here as well; with $a_i \in \{0, 1\}$ its numerator is $(r + \lambda) V_i' + 2\pi V_i''$. Lemma A.3 gives $V_i' \geq 0$ and Lemma 2 gives $V_i'' < 0$ wherever these derivatives exist. At any such state where $G_i \geq 0$, dividing by $-V_i'' > 0$ and using $\frac{\pi}{\sigma} < \frac{\sqrt{r+\lambda}}{2}$ gives

$$\frac{V_i'}{-V_i''} \geq \frac{\frac{1}{2} \sigma^2}{\pi} > \frac{\pi}{\frac{1}{2}(r + \lambda)},$$

so $(r + \lambda) V_i' + 2\pi V_i'' > 0$ and hence

$$G_i' > 0 \quad \text{wherever} \quad G_i \geq 0. \quad (\text{A.16})$$

Since $G_i(x) \geq 0$, it follows from (A.16) that $G_i > 0$ at all higher states where V_i'' exists.¹⁵ Hence, the best-response condition (A.2) gives $s_i^*(\Delta k) = 1$ wherever V_i'' exists.

This completes the proof. $\square$

Proof of Propositions 2' and 3'. Assume that $\pi > 0$, and that the agents play according to some well-behaved MPE $(k_i^*, s_i^*, k_j^*, s_j^*)$.

By Lemma 1, the follower takes the risky move throughout his continuation region. We first characterize the leader's action rule. Define $G_i \equiv \pi V_i' + \frac{1}{2} \sigma^2 V_i''$. If $\frac{\pi}{\sigma} > \frac{\sqrt{r+\lambda}}{2}$, Lemma A.4 implies that the leader takes the risky move wherever V_i'' exists on $(0, k_j^*)$. If $\frac{\pi}{\sigma} < \frac{\sqrt{r+\lambda}}{2}$, by Lemma A.5, the leader's risky region is upward closed. In either case, setting

$$k_i^{**} \equiv \inf \{x \in (0, k_j^*) : G_i(x) > 0\},$$

¹⁵Were G_i to return to zero at some higher state, the smallest such state x_0 would have $G_i > 0$ on (x, x_0) and $G_i(x_0) = 0$, so $G_i'(x_0) \leq 0$, contradicting (A.16); the sign of G_i is preserved across switching states as in the proof of Lemma A.4, so the conclusion holds on all of (x, k_j^*) wherever these derivatives exist.

the leader takes the safe action on $(0, k_i^{**})$ and the risky action on (k_i^{**}, k_j^*) , up to payoff-irrelevant nonregular points.

Given agent j 's choice of stopping boundary k_j^* and switching point k_j^{**} , and given candidate best-response boundaries k_i^* and k_i^{**} for agent i , the HJB equation (A.1) becomes

$$rV_i = \begin{cases} \lambda(P - V_i) - c + \sigma^2 V_i'', & k_i^{**} < \Delta k < k_j^*, \\ \lambda(P - V_i) - c - \pi V_i' + \frac{1}{2}\sigma^2 V_i'', & 0 < \Delta k < k_i^{**}, \\ \lambda(0 - V_i) - c + \pi V_i' + \frac{1}{2}\sigma^2 V_i'', & -k_j^{**} < \Delta k < 0, \\ \lambda(0 - V_i) - c + \sigma^2 V_i'', & -k_i^* < \Delta k < -k_j^{**}. \end{cases}$$

Solving these second-order linear ODEs subject to value matching (A.3) at k_j^* , value matching and smooth pasting (A.4) at $-k_i^*$, continuity of V_i and V_i' at the tie (A.6) and at the switching thresholds k_i^{**} and $-k_j^{**}$, and the leader's indifference condition at the switch,

$$\pi V_i'(k_i^{**}) + \frac{1}{2}\sigma^2 V_i''(k_i^{**}) = 0, \quad (\text{A.17})$$

we obtain

$$V_i(\Delta k) = \begin{cases} \frac{\lambda P - c}{r + \lambda} + C_{1i}e^{-\eta\Delta k} + C_{2i}e^{\eta\Delta k}, & k_i^{**} < \Delta k < k_j^*, \\ \frac{\lambda P - c}{r + \lambda} + C_{3i}e^{-\xi_+\Delta k} + C_{4i}e^{-\xi_-\Delta k}, & 0 < \Delta k < k_i^{**}, \\ -\frac{c}{r + \lambda} + C_{5i}e^{\xi_+\Delta k} + C_{6i}e^{\xi_-\Delta k}, & -k_j^{**} < \Delta k < 0, \\ -\frac{c}{r + \lambda} + C_{7i}e^{\eta\Delta k} + C_{8i}e^{-\eta\Delta k}, & -k_i^* < \Delta k < -k_j^{**}, \end{cases}$$

where $\xi_+ > 0$, $\xi_- < 0$, and $\eta > 0$ are the roots from (A.7) and (A.9). These conditions determine the eight constants and, through smooth pasting and the indifference condition, the best-response boundaries k_i^* and k_i^{**} , taking k_j^* and k_j^{**} as given. In particular, the indifference condition (A.17) determines any interior switching point k_i^{**} through

$$\frac{e^{\eta(k_j^* - k_i^{**})} - e^{-\eta(k_j^* - k_i^{**})}}{e^{\eta(k_j^* - k_i^{**})} + e^{-\eta(k_j^* - k_i^{**})}} = \frac{2\pi}{\eta\sigma^2} \implies k_i^{**} = k_j^* - \frac{1}{2\eta} \log \frac{1 + \frac{2\pi}{\eta\sigma^2}}{1 - \frac{2\pi}{\eta\sigma^2}},$$

valid whenever $0 < \frac{2\pi}{\eta\sigma^2} < 1$. Thus, agent i best responds with

$$k_i^{**} = \max \left\{ 0, k_j^* - \frac{1}{2\eta} \log \frac{1 + \frac{2\pi}{\eta\sigma^2}}{1 - \frac{2\pi}{\eta\sigma^2}} \right\}. \quad (\text{A.18})$$

If $\frac{2\pi}{\eta\sigma^2} \geq 1$, then Lemma A.4 implies that the leader takes the risky move throughout, so $k_i^{**} = 0$.

Analogous to the proof of Proposition 1', for fixed k_j^* (and thus k_i^{**}) the stopping boundary $k_i^* = \widehat{BR}(k_j^*)$ is the unique positive solution to the smooth-pasting condition (A.4) at $-k_i^*$, with $\widehat{BR}' \in (-1, 0)$ and limiting values in $(0, \infty)$. Defining $\widehat{Z}(k) \equiv \widehat{BR}(\widehat{BR}(k)) - k$, we have $\widehat{Z}'(k) = \widehat{BR}'(\widehat{BR}(k))\widehat{BR}'(k) - 1 < 0$, with $\widehat{Z}(0) > 0$ and $\lim_{k \rightarrow \infty} \widehat{Z}(k) < 0$, so there is a unique $k^* \in (0, \infty)$ with $k^* = \widehat{BR}(k^*)$. By symmetry $k_i^* = k_j^* = k^*$ and $k_i^{**} = k_j^{**} = k^{**}$. The switching point k^{**} is interior if and only if $k^* > \frac{1}{2\eta} \log \frac{1+2\pi/(\eta\sigma^2)}{1-2\pi/(\eta\sigma^2)}$, which by

(A.18) holds exactly when

$$\frac{\pi}{\sigma} < \frac{\sqrt{r+\lambda}}{2} \cdot \frac{\sqrt{2\phi(\sqrt{\phi^2+8}+\phi)}-8}{\sqrt{\phi^2+8}+\phi} = \frac{\sqrt{r+\lambda}}{2} f(\phi).$$

Hence $k^{**} > 0$ in the medium return case and $k^{**} = 0$ in the high return case.

Substituting $k_i^* = k_j^* = k^*$ and $k_i^{**} = k_j^{**} = k^{**}$ into the best-response system simplifies it to the symmetric stopping boundary. Propositions 2 and 3 then verify that the symmetric strategy profile with this boundary is an MPE, with $k^{**} = 0$ in the high return case and $0 < k^{**} < k^*$ in the medium return case. Therefore, the high and medium return cases each have a unique well-behaved MPE, and it is symmetric.

This completes the proof. $\square$

A.2 Proofs for Section 4

Proof of Proposition 4. In each of the three types of equilibria, let $\tau_{k^*} \equiv \inf \{t \geq 0 : |k_i(t) - k_j(t)| \geq k^*\}$ denote the first hitting time to the stopping boundary, and let

$$\Psi(\theta; k^*, k_0) \equiv \mathbb{E} \left[e^{-\theta \tau_{k^*}} \mid k_i(0) - k_j(0) = k_0 \right], \quad \theta > 0$$

denote its Laplace transform.

Let τ_S be the success time. Conditional on the realized progress path, success arrives at rate $\bar{\lambda} + \underline{\lambda}$ while both agents stay in the contest, and drops to $\bar{\lambda}$ once the follower has dropped out at τ_{k^*} .

We first compute the expected length of time during which both agents stay in the contest, $\mathbb{E}[\tau_{k^*} \wedge \tau_S \mid k_0]$. For a fixed realization $\tau_{k^*} = x$,

$$\begin{aligned} \mathbb{E}[\tau_{k^*} \wedge \tau_S \mid \tau_{k^*} = x] &= \int_0^\infty \mathbb{P}(\tau_{k^*} \wedge \tau_S > t \mid \tau_{k^*} = x) dt \\ &= \int_0^x \mathbb{P}(\tau_S > t \mid \tau_{k^*} = x) dt \\ &= \int_0^x e^{-(\bar{\lambda} + \underline{\lambda})t} dt = \frac{1 - e^{-(\bar{\lambda} + \underline{\lambda})x}}{\bar{\lambda} + \underline{\lambda}}. \end{aligned}$$

Taking expectations over τ_{k^*} gives

$$\mathbb{E}[\tau_{k^*} \wedge \tau_S \mid k_0] = \frac{1 - \mathbb{E} \left[e^{-(\bar{\lambda} + \underline{\lambda})\tau_{k^*}} \mid k_0 \right]}{\bar{\lambda} + \underline{\lambda}} = \frac{1 - \Psi(\bar{\lambda} + \underline{\lambda}; k^*, k_0)}{\bar{\lambda} + \underline{\lambda}}.$$

Therefore, Objectives 2 and 3 are equivalent as long as Ψ is strictly decreasing in k^* .

To show that Ψ is strictly decreasing in k^* , first note that $k_i - k_j$ has almost surely continuous sample paths in all three types of equilibria. Therefore, τ_{k^*} is strictly increasing in k^* in the sense of first-order stochastic dominance. Next, observe that $e^{-\theta x}$ is strictly decreasing in x . Combining this with the monotonicity of τ_{k^*} , we conclude that $\Psi(\theta; k^*, k_0) = \mathbb{E} \left[e^{-\theta \tau_{k^*}} \mid k_i(0) - k_j(0) = k_0 \right]$ is strictly decreasing in k^* .¹⁶

Next consider the expected time needed to achieve success, $\mathbb{E}[\tau_S \mid k_0]$. For a fixed realization $\tau_{k^*} = x$,

$$\mathbb{P}(\tau_S > t \mid \tau_{k^*} = x) = \begin{cases} e^{-(\bar{\lambda} + \underline{\lambda})t}, & t < x, \\ e^{-(\bar{\lambda} + \underline{\lambda})x} e^{-\bar{\lambda}(t-x)}, & t \geq x. \end{cases}$$

¹⁶As a sanity check, in the low return case ($\pi \leq 0$), $|\Delta k|$ follows a reflected Brownian motion with drift $-\pi$ and volatility σ . It

Hence,

$$\begin{aligned}
\mathbb{E}[\tau_S \mid \tau_{k^*} = x] &= \int_0^\infty \mathbb{P}(\tau_S > t \mid \tau_{k^*} = x) dt \\
&= \int_0^x e^{-(\bar{\lambda} + \underline{\lambda})t} dt + \int_x^\infty e^{-(\bar{\lambda} + \underline{\lambda})x} e^{-\bar{\lambda}(t-x)} dt \\
&= \frac{1 - e^{-(\bar{\lambda} + \underline{\lambda})x}}{\bar{\lambda} + \underline{\lambda}} + \frac{e^{-(\bar{\lambda} + \underline{\lambda})x}}{\bar{\lambda}}.
\end{aligned}$$

Taking expectations over τ_{k^*} gives

$$\begin{aligned}
\mathbb{E}[\tau_S \mid k_0] &= \frac{1 - \Psi(\bar{\lambda} + \underline{\lambda}; k^*, k_0)}{\bar{\lambda} + \underline{\lambda}} + \frac{\Psi(\bar{\lambda} + \underline{\lambda}; k^*, k_0)}{\bar{\lambda}} \\
&= \frac{1}{\bar{\lambda} + \underline{\lambda}} + \frac{\underline{\lambda}}{\bar{\lambda}(\bar{\lambda} + \underline{\lambda})} \Psi(\bar{\lambda} + \underline{\lambda}; k^*, k_0).
\end{aligned}$$

The monotonicity of Ψ in k^* indicates that Objectives 1 and 3 are also equivalent.

This completes the proof. $\square$

Proof of Proposition 5. By Proposition 4, the designer's problem reduces to maximizing k^* over all prize pairs $(\bar{P}, \underline{P})$ with the budget constraint $\bar{P} + \underline{P} \leq B$.

1. If $(\bar{\lambda} + \underline{\lambda})B/2 < c$, then for all choices of $(\bar{P}, \underline{P})$ the denominator of $\bar{\phi}$, $c - (\underline{\lambda}\bar{P} + \bar{\lambda}\underline{P})$, is strictly positive. It follows that

$$\begin{aligned}
\frac{(\bar{\lambda} - \underline{\lambda})(\bar{P} - \underline{P})}{c - (\underline{\lambda}\bar{P} + \bar{\lambda}\underline{P})} &\leq \frac{(\bar{\lambda} - \underline{\lambda})B}{c - \underline{\lambda}B} \iff cB - c\bar{P} + (c - \bar{\lambda}B - \underline{\lambda}B)\underline{P} \geq 0 \\
&\iff cB - c(B - \underline{P}) + (c - \bar{\lambda}B - \underline{\lambda}B)\underline{P} \geq 0 \\
&\iff (c - (\bar{\lambda} + \underline{\lambda})B/2)\underline{P} \geq 0,
\end{aligned}$$

as desired. Equality holds if $\bar{P} = B$ and $\underline{P} = 0$.

2. If $(\bar{\lambda} + \underline{\lambda})B/2 > c$, then the denominator of $\bar{\phi}$, $c - (\underline{\lambda}\bar{P} + \bar{\lambda}\underline{P})$, can be negative if $\bar{P} = B/2 + \varepsilon$ and $\underline{P} = B/2 - \varepsilon$, with $\varepsilon \in (0, ((\bar{\lambda} + \underline{\lambda})B/2 - c) / (\bar{\lambda} - \underline{\lambda}))$. For these prizes the flow payoff from staying in the contest is positive at every position, so neither agent drops out and $k^* = \infty$, which is optimal.

This completes the proof. $\square$

follows from Theorem 1 in [Mayerhofer \(2019\)](#) that

$$\Psi(\theta; k^*, k_0) = e^{\frac{(-\pi)(k^* - k_0)}{\sigma^2}} \frac{\sqrt{\pi^2 + 2\theta\sigma^2} \cosh\left(\frac{k_0 \sqrt{\pi^2 + 2\theta\sigma^2}}{\sigma^2}\right) + (-\pi) \sinh\left(\frac{k_0 \sqrt{\pi^2 + 2\theta\sigma^2}}{\sigma^2}\right)}{\sqrt{\pi^2 + 2\theta\sigma^2} \cosh\left(\frac{k^* \sqrt{\pi^2 + 2\theta\sigma^2}}{\sigma^2}\right) + (-\pi) \sinh\left(\frac{k^* \sqrt{\pi^2 + 2\theta\sigma^2}}{\sigma^2}\right)},$$

which is indeed strictly decreasing in k^* .